
Chapter 2

Basic MOS Device Physics

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Design of Analog CMOS Integrated Circuits

--Ch.2 Basic MOS Device Physics # 1

Outline

- General consideration
- MOS I/V characteristics
- MOS small signal model
- Second-order effects



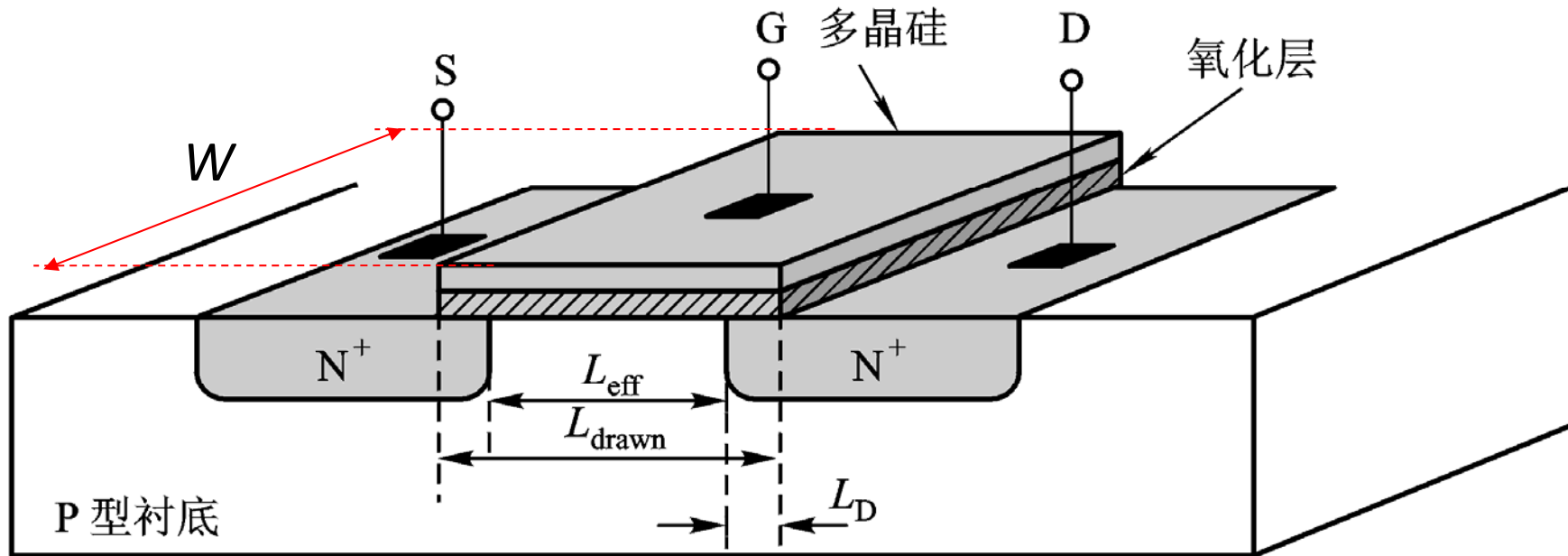
Outline

- **General consideration**
 - MOS physical structure
 - MOS symbols
- **MOS I/V characteristics**
- **MOS small signal model**
- **Second-order effects**



MOSFET Structure

An **NMOS** on a p-type substrate (or p “body”, “bulk”)



design parameters:

L_{drawn} : length

W : width

Process parameters:

t_{ox} : thickness of oxide

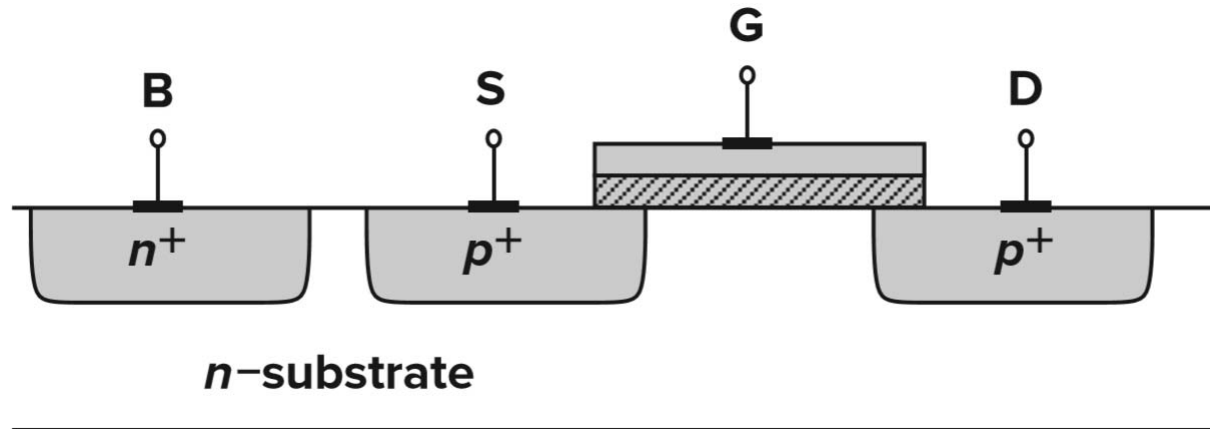
N_{sub} : doping concentration of sub.



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Substrate potential

- Sub potential influences MOS characteristics

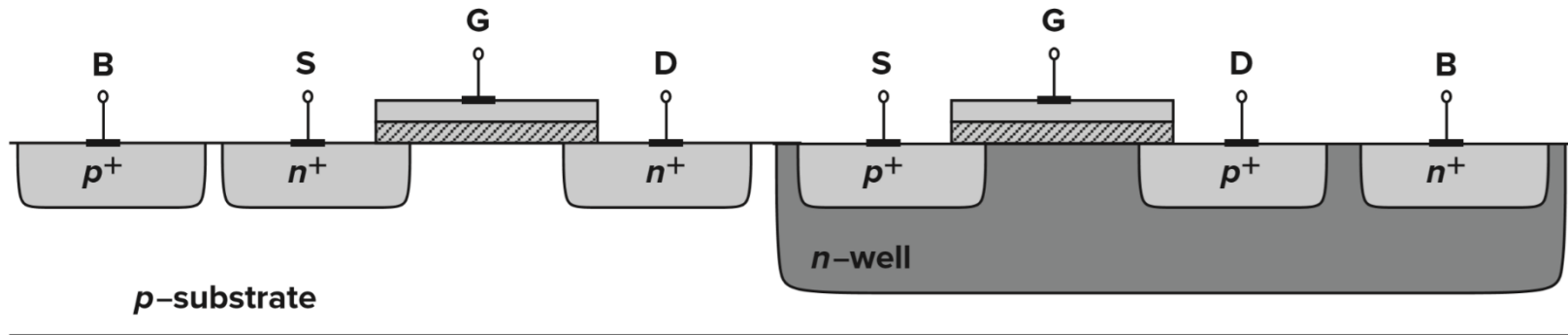


- MOS is essentially a 4-terminal device
 - G: Gate
 - S: Source
 - D: Drain
 - B: Body



Obtain NMOS and PMOS on a p-sub

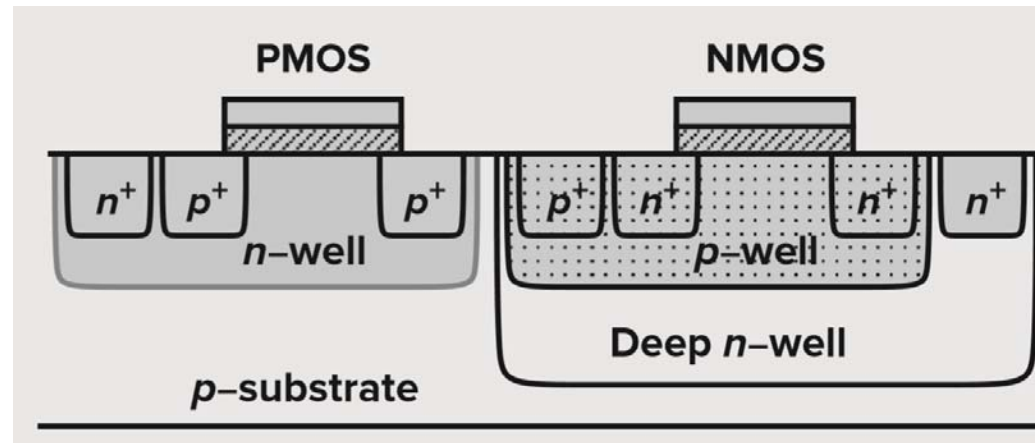
- Use N-well to fabricate PMOS



- We call this double-well CMOS process
 - **Question:** if we have 2 NMOS on this wafer, can we connect the B terminals of them to different potentials? Why?
 - Answer the same question for PMOS.



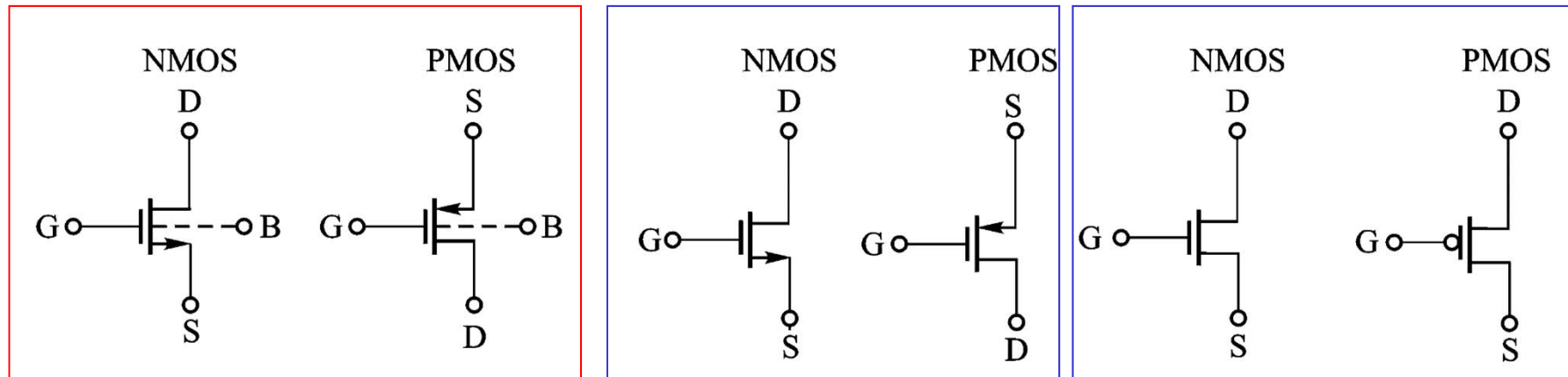
NMOS in a Deep N-well



- A solution to realize NMOS transistors with separate bulks
- Advantages:
 - Can realize independent B for each NMOS
 - Improved performances due to reduced cross-talks
- Disadvantages
 - Larger circuit area (layout overhead)
 - Higher cost (additional mask)



MOS symbols



Sub potentials must be considered for analog IC

4-terminal symbols are often adopted

In digital IC:

NMOS B → GND

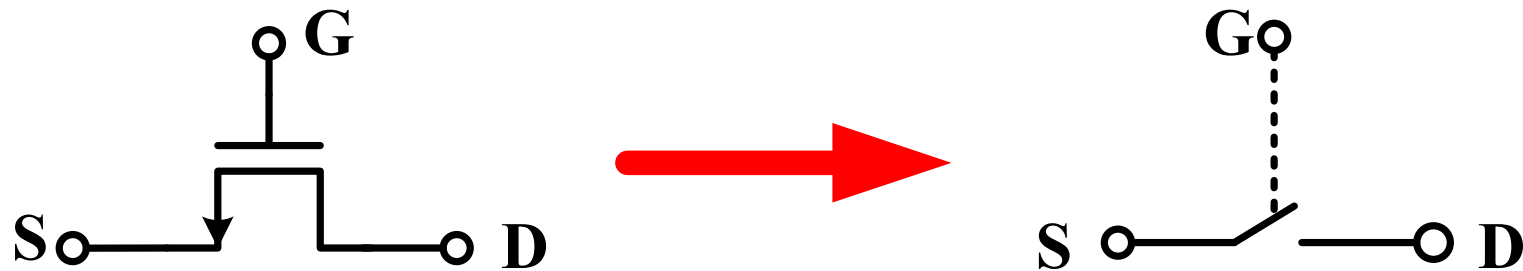
PMOS B → VDD

Use 3-terminal symbols



MOS as a switch

- MOS transistor can be thought as a switch
 - Digital IC
 - Control transistors in **analog IC**

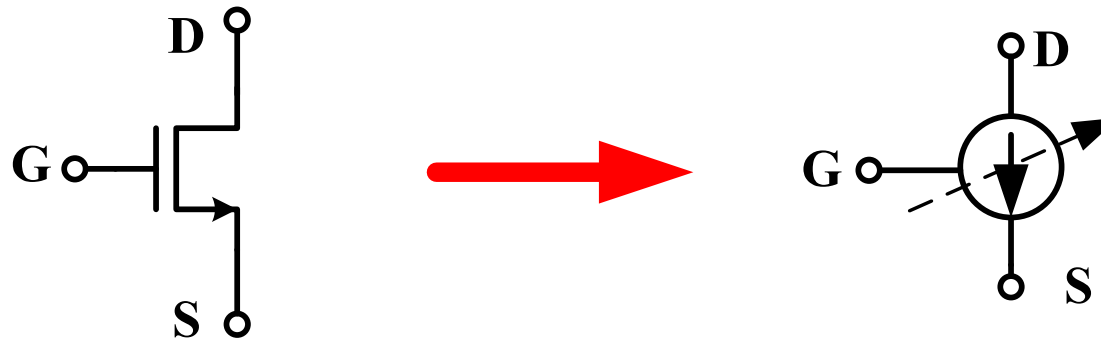


- Question: how to turn on (or off) the MOS ?
 - Need to know the threshold voltage (V_{th})



MOS as a VCCS

- MOS is also a **voltage controlled current source (VCCS)**
 - In analog circuits



- Question: how does the gate voltage control the drain/source current?
 - Need to derive the I/V characteristics

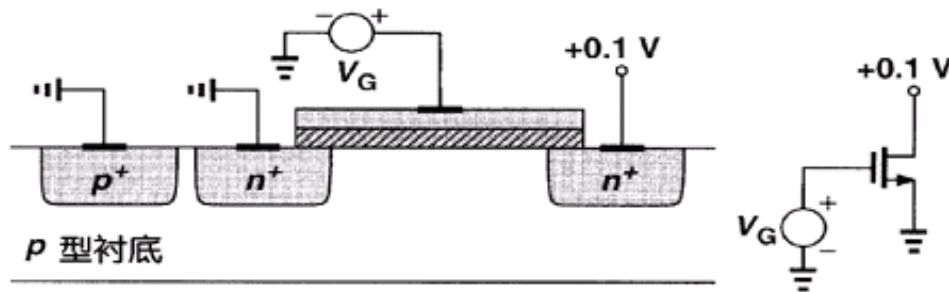


Outline

- General consideration
- **MOS I/V characteristics**
 - Threshold Voltage
 - I/V characteristics derivation
- MOS small signal model
- Second-order effects



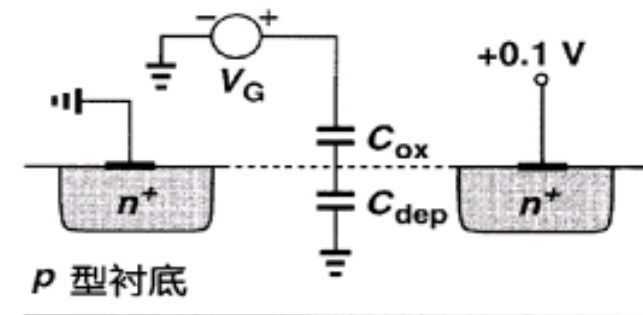
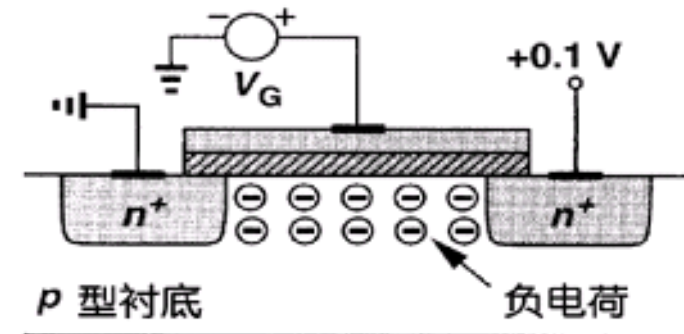
Turn-on process for an NMOS

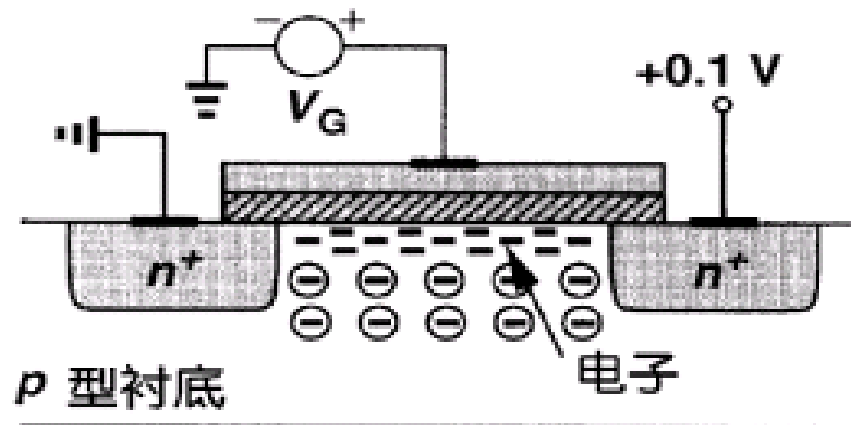


When $V_G=0$, MOS is off, no current

V_G increases a bit, surface is depleted.
Still off, still no current.

Width of depletion region and surface potential increases with V_G , the structure resembles two caps in series.





When V_G increases to a sufficiently positive value, the surface is “inverted”, the MOS is turned on. The value of V_G at this point is called V_{TH}

If V_G rises further, the charge density in channel continues to increase (I_D increases).



Threshold Voltage Definition

- V_{TH} : the gate voltage when the surface is inverted
- **Inversion definition:** electron concentration in the channel equals to the hole concentration in substrate

$$V_{TH} = \Phi_{MS} + 2\Phi_F + \frac{Q_{dep}}{C_{ox}}$$

Φ_{MS} : Work function difference between the gate and substrate

Φ_F : Fermi level

Q_{dep} : Charge density in the depletion region

C_{ox} : Gate oxide capacitance per unit area

See Chapter 8, "Semiconductor Physics" for detailed derivation



Important note about V_{TH}

$$V_{TH} = \Phi_{MS} + 2\Phi_F + \frac{Q_{dep}}{C_{ox}}$$

$$\Phi_F = (kT / q) \ln(N_{sub} / n_i) \quad Q_{dep} = \sqrt{4q\epsilon_{si} |\Phi_F| N_{sub}}$$

For NMOS:

the higher the substrate doping, the higher the V_{TH}

For a given process under given temperature, V_{TH} is almost a constant.

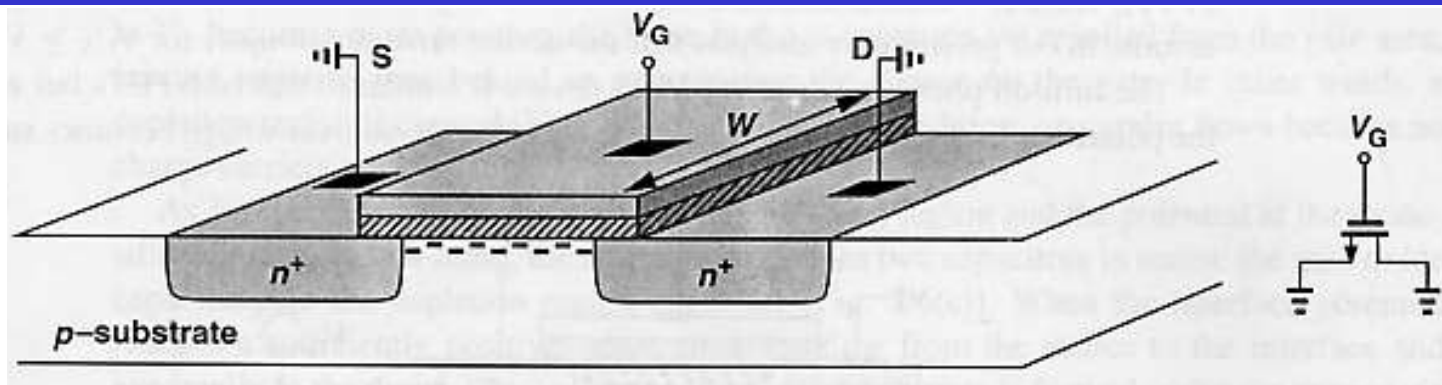


Outline

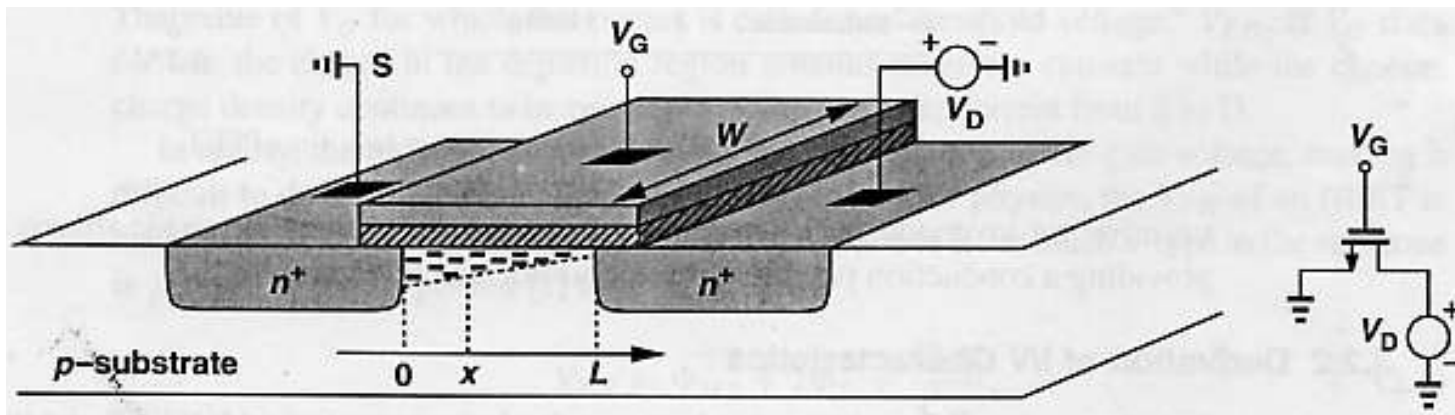
- General consideration
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Derivation of I/V Characteristics



$V_{GS} > V_{TH}$, inverted, but $V_D = V_S$, no current flows



$V_D > V_S$, current flows from D to S

How to obtain I_{DS} as the *functions* of the terminal voltages?



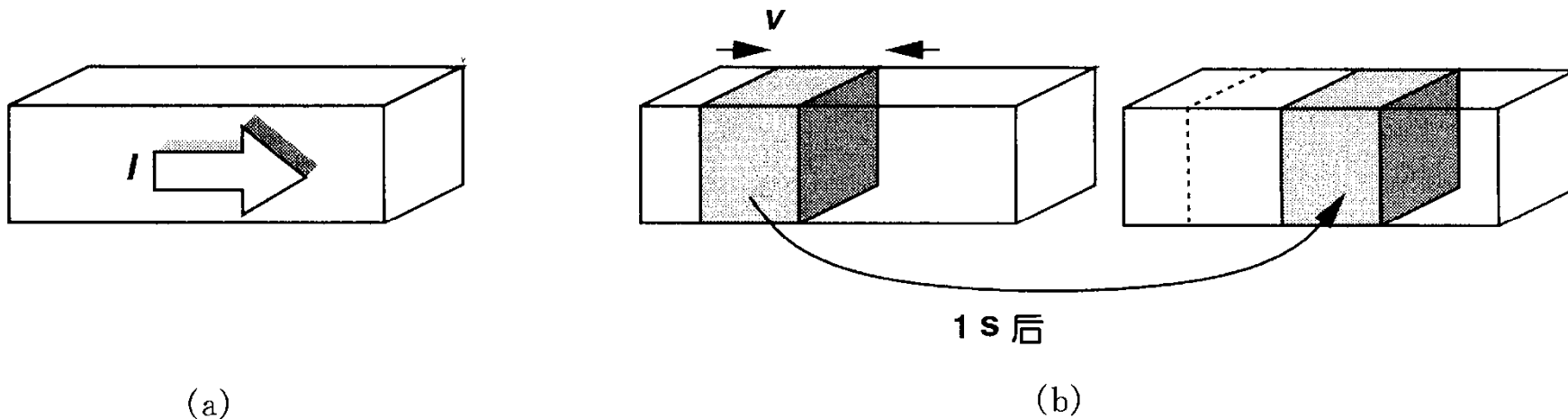
Derivation of I/V Characteristics

1st observation:

Charge density along the current direction is Q_d (C/m)

Charge velocity is v (m/s)

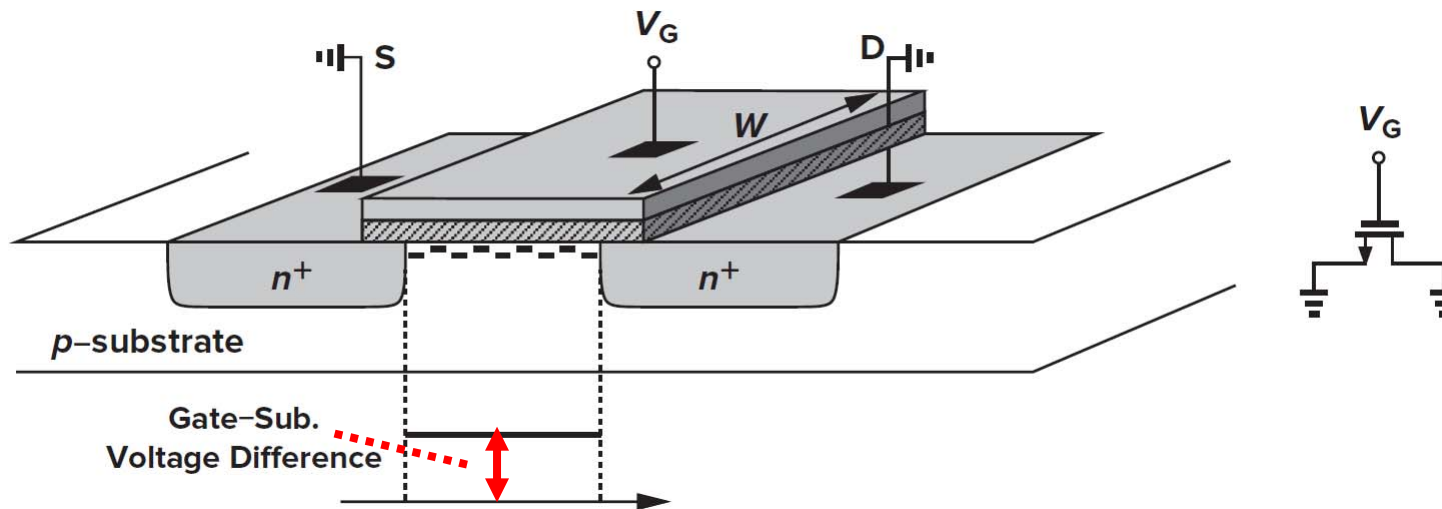
$$I = Q_d \cdot v$$



- **2nd observation**

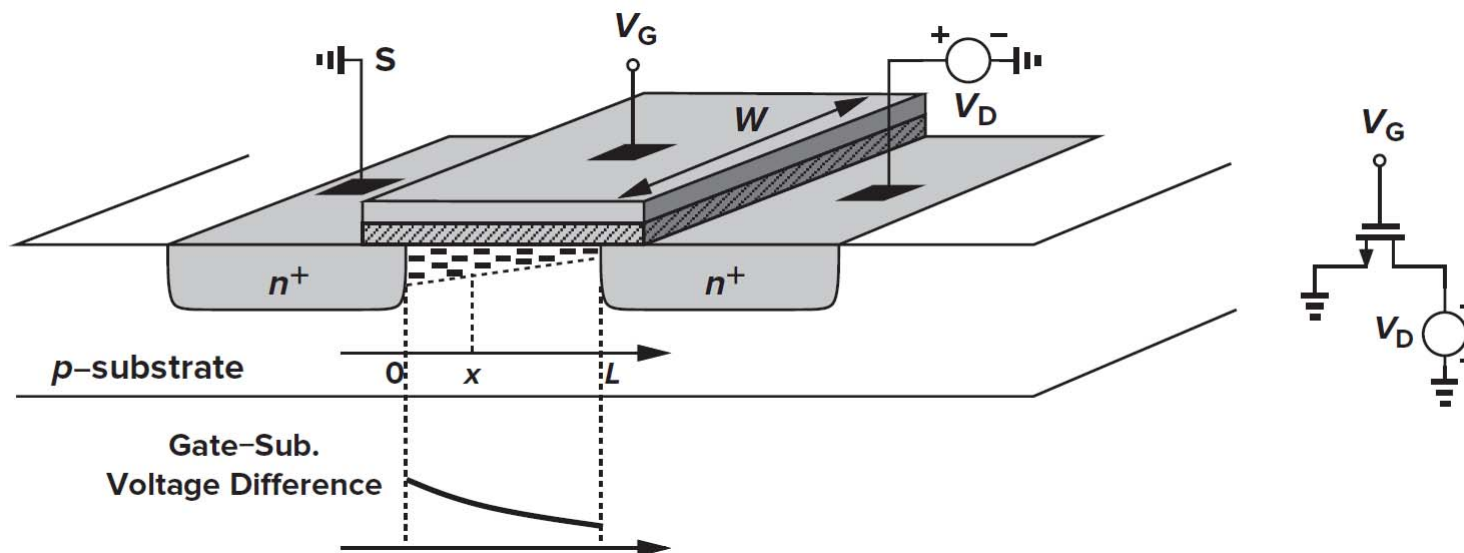
- Assume $V_{GS} > V_{TH}$, and S and D are all connected to ground,
- Uniform charge density is obtained

$$Q_d = WC_{ox}(V_{GS} - V_{TH})$$



- Then suppose V_D is a little bit larger than V_S
- Charge density is no longer distributed uniformly, but a function of position x

$$Q_d(x) = WC_{OX} [V_{GS} - V(x) - V_{TH}]$$



$$Q_d(x) = WC_{ox}[V_{GS} - V(x) - V_{TH}]$$

$$I_D = Q_d \cdot v = -WC_{ox}[V_{GS} - V(x) - V_{TH}] \cdot v$$

$$\text{Given } v = \mu_n E, \quad E(x) = -\frac{dV(x)}{dx}$$

$$I_D = WC_{ox}[V_{GS} - V(x) - V_{TH}]\mu_n \frac{dV(x)}{dx}$$



$$I_D = WC_{ox} [V_{GS} - V(x) - V_{TH}] \mu_n \frac{dV(x)}{dx}$$

- Considering boundary conditions: $V(0)=0$, $V(L) = V_{DS}$
- Multiplying both sides with dx , and perform integration

$$\int_{x=0}^L I_D dx = \int_{V=0}^{V_{DS}} WC_{ox} \mu_n [V_{GS} - V(x) - V_{TH}] dV$$

$$I_D = \mu_n C_{ox} \frac{W}{L} [(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2]$$



More simple method

$$I_D = Q_d \cdot v$$

- Use average Q_d (at $L/2$), and average electrical field

$$I_D = WC_{ox} [V_{GS} - V(x)|_{x=L/2} - V_{TH}] \cdot \mu_n \bar{E}$$

$$= WC_{ox} (V_{GS} - \frac{1}{2} V_{DS} - V_{TH}) \cdot \mu_n \frac{V_{DS}}{L}$$

$$= \frac{W}{L} \mu_n C_{ox} [(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2]$$



I/V Characteristics in “triode region”

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{V_{DS}^2}{2} \right], \quad \text{for } V_{DS} \leq (V_{GS} - V_{TH})$$

If $V_{DS} \leq V_{GS} - V_{TH}$, we define:

NMOS operates in “triode region” or “linear region”

In triode region:

I_D is influenced both by V_{GS} and V_{DS}

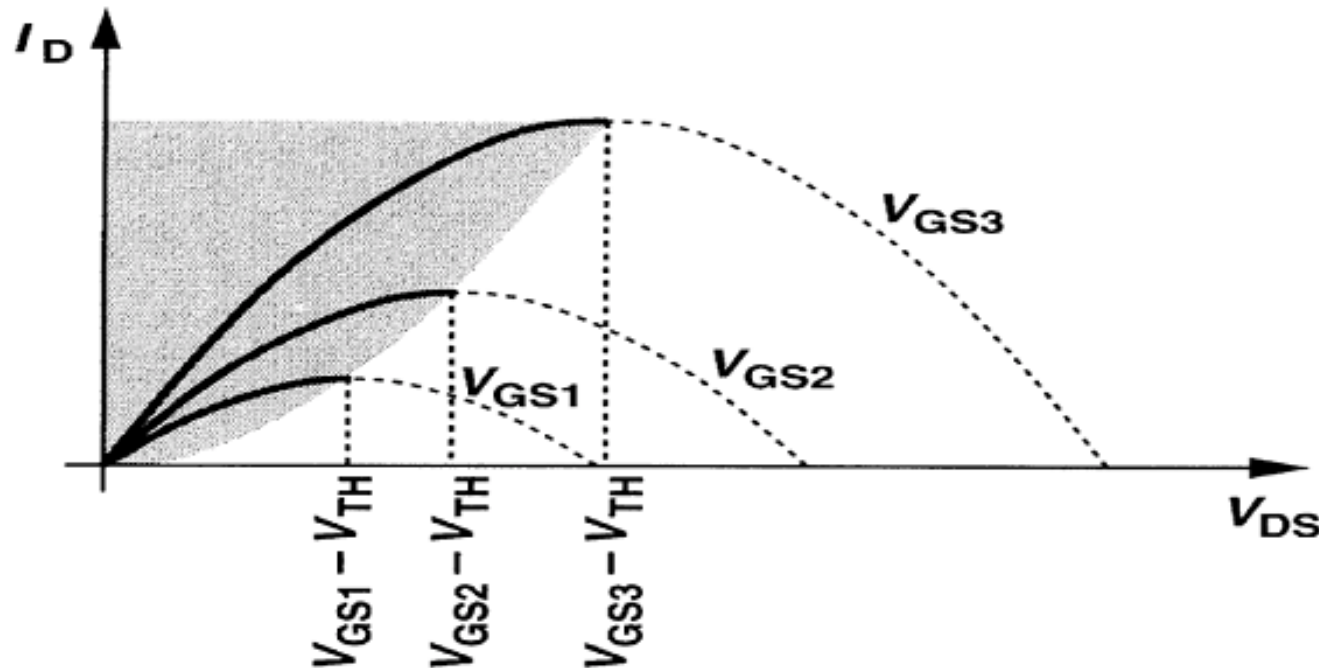
W/L is called “aspect ratio”, a design parameter

$\mu_n C_{ox}$ is a process parameter



I/V Characteristics in “triode region”

$$I_D = \mu_n C_{ox} \frac{W}{L} [(V_{GS} - V_{TH}) V_{DS} - \frac{V_{DS}^2}{2}], \quad \text{for } V_{DS} \leq (V_{GS} - V_{TH})$$



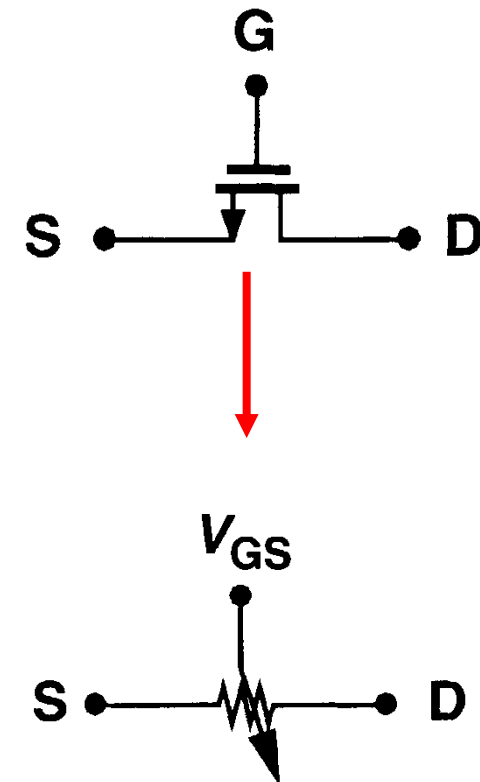
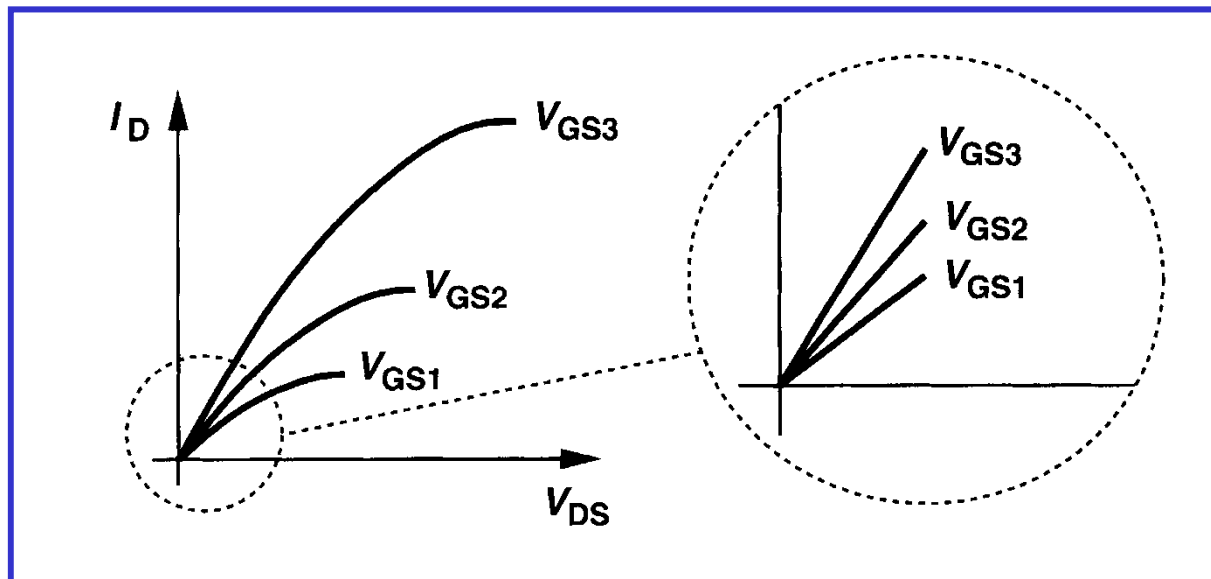
$$I_{D,\max} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} [(V_{GS} - V_{TH})^2], \quad \text{when } V_{DS} = V_{GS} - V_{TH}$$



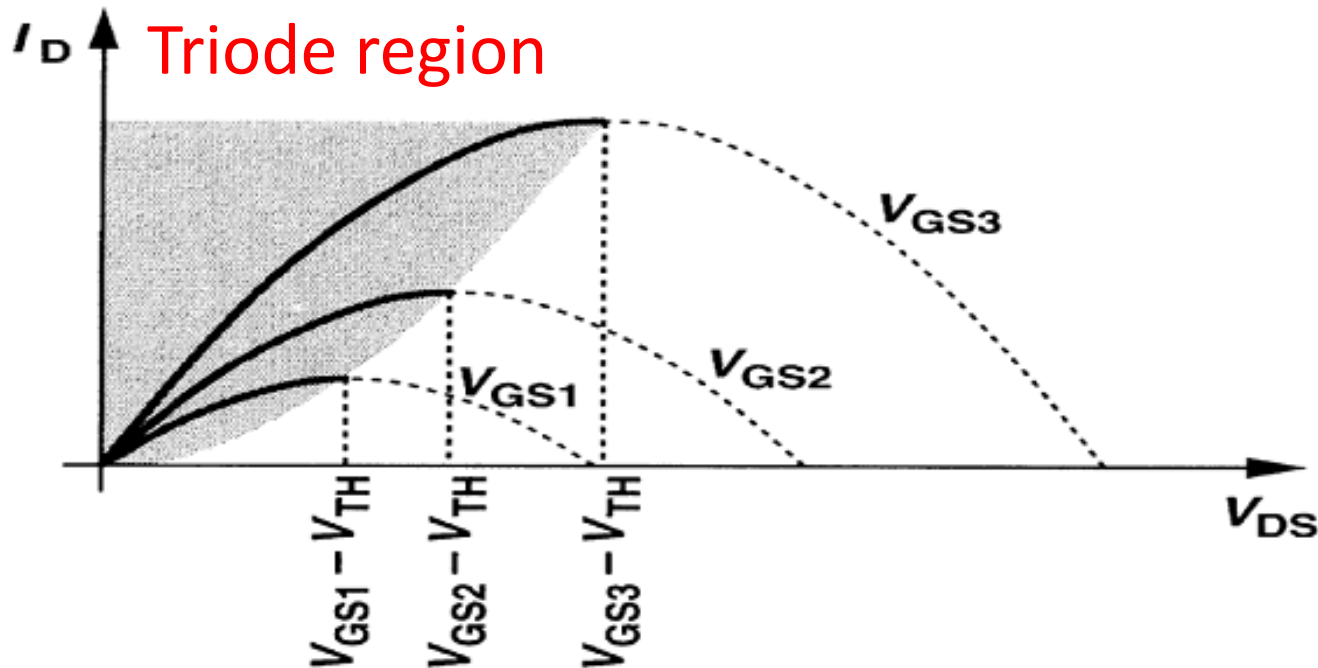
Application of NMOS in Triode Region

$$I_D \approx \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH}) V_{DS} \quad \text{if } \frac{1}{2} V_{DS} \ll V_{GS} - V_{TH}$$

→ $R_{on} = \frac{V_{DS}}{I_D} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})}$



I/V Characteristics in Saturation Region



- What happens if $V_{DS} > V_{GS} - V_{TH}$?
- Will it follow the parabolic curve?



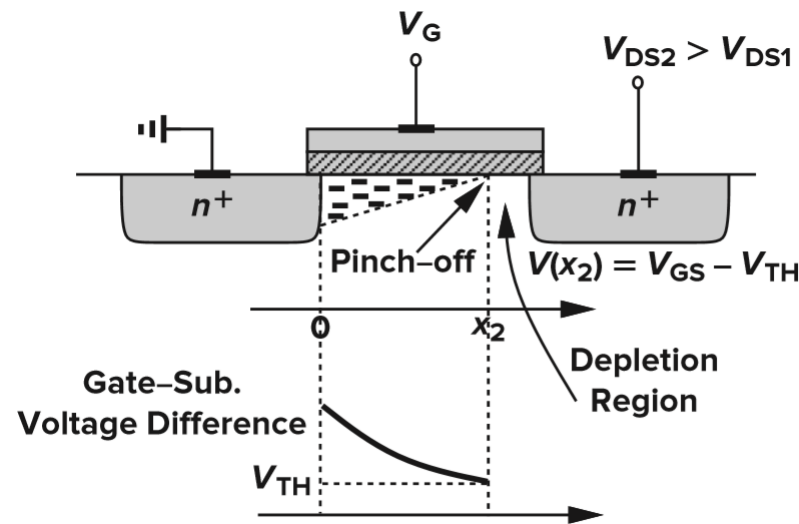
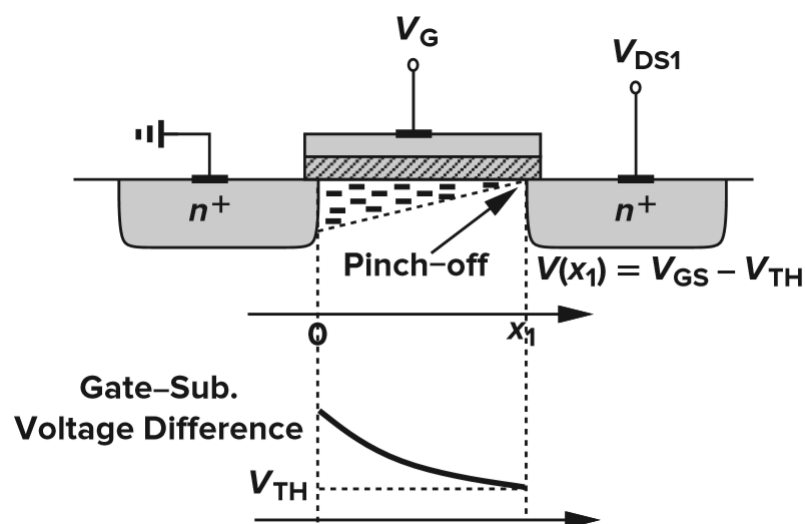
Operation in Saturation Region

Recall that: $Q_d(x) = WC_{OX}[V_{GS} - V(x) - V_{TH}]$

$Q_d(x)$ drops to **0**, if $V(x)$ approaches $V_{GS} - V_{TH}$

So, if V_{DS} is slightly greater than $V_{GS} - V_{TH}$, “pinch off” happens at x_1

Pinch off point moves to left when V_{DS} increases



Saturation Region ($V_{DS} > V_{GS} - V_{TH}$)

$$I_D = \mu_n C_{ox} \frac{W}{L'} [(V_{GS} - V_{TH}) V'_{DS} - \frac{1}{2} V'^2_{DS}]$$

$$V'_{DS} = V_{GS} - V_{TH}, \quad (\text{pinch-off})$$

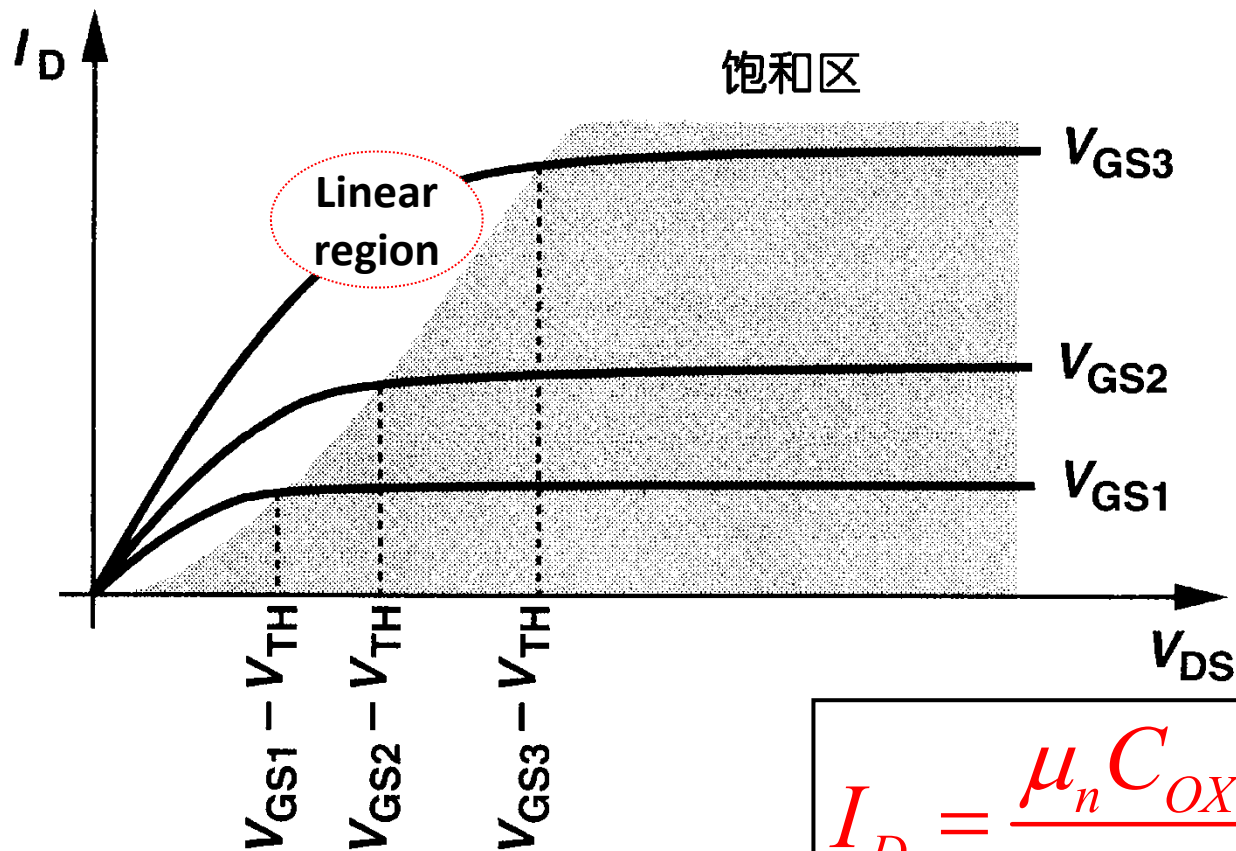
$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L'} (V_{GS} - V_{TH})^2$$

For long channel MOS, $L' \approx L$

→
$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 \text{ for } V_{DS} > V_{GS} - V_{TH}$$



Saturation Region



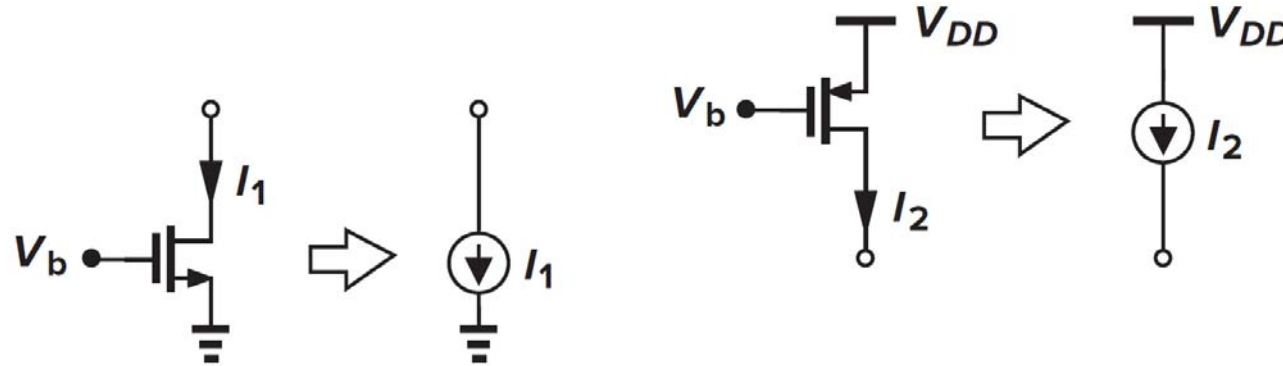
$$I_D = \frac{\mu_n C_{OX}}{2} \frac{W}{L} (V_{GS} - V_{TH})^2$$

I_D is controlled by V_{GS} , independent of V_{DS} , “saturated”



Application of MOS in Saturation

- Saturated MOS can be used as current source



$$I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{GS} - V_{TH})^2 = \frac{\mu_n C_{ox}}{2} \frac{W}{L} \cdot V_{OD}^2$$

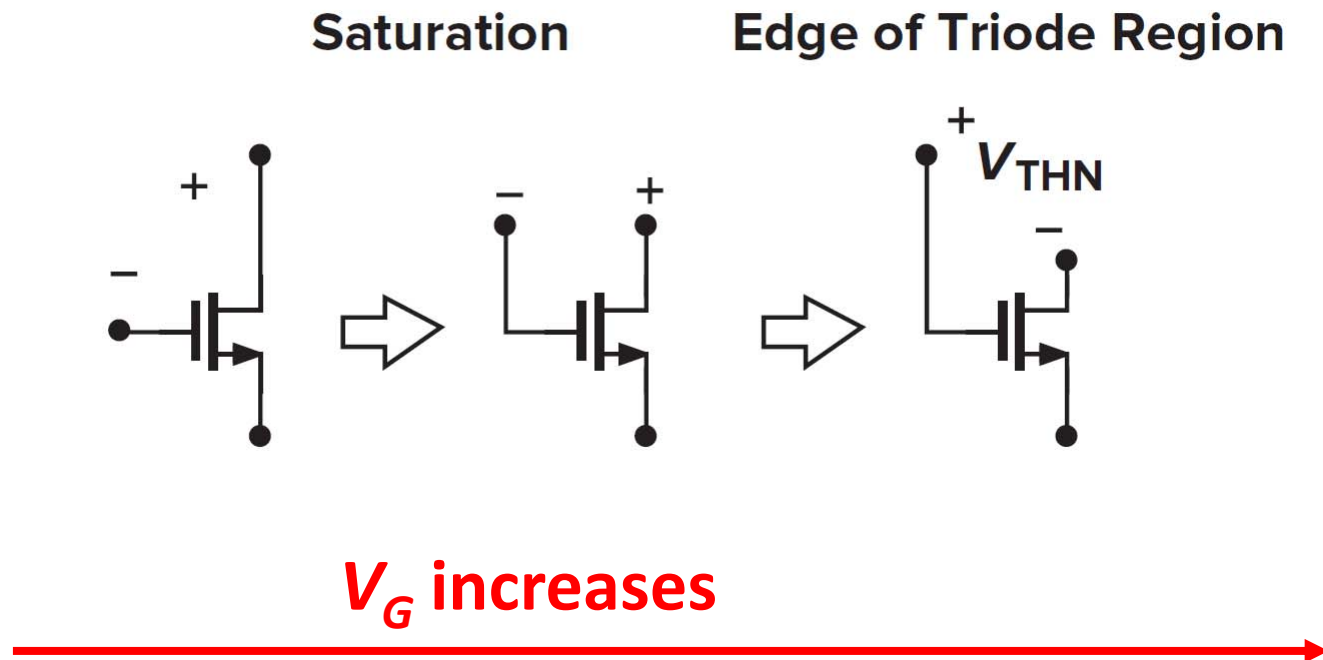
$V_{OD} = V_{GS} - V_{TH}$ is called “overdrive voltage”

$$V_{OD} = V_{GS} - V_{TH} = \sqrt{\frac{2I_D}{\mu_n C_{ox} (W/L)}}$$



Operation region changes with V_G

- For NMOS, if $V_{GS} < V_{TH}$, it is in off state.
- When V_{GS} increases, it operates from saturation region to triode region.



NMOS I/V characteristics summary

- Off: $V_{GS} - V_{TH} \leq 0$

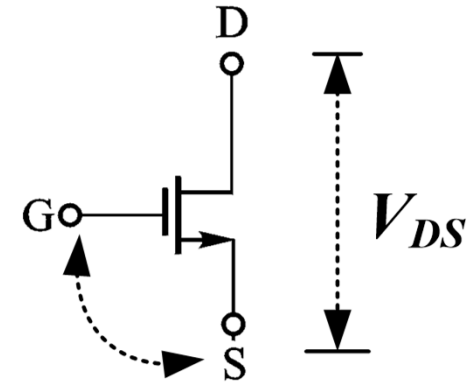
$$I_{DS} = 0$$

- Triode region: $0 < V_{DS} \leq V_{GS} - V_{TH}$

$$I_{DS} = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

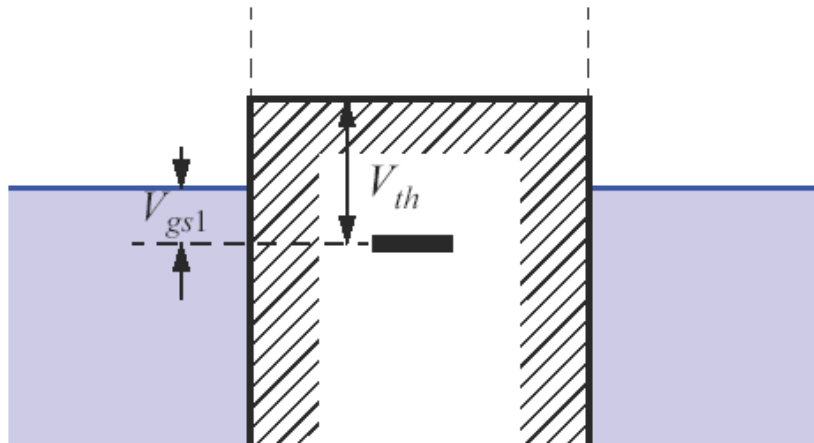
- Saturation region: $0 < V_{GS} - V_{TH} < V_{DS}$

$$I_{DS} = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{GS} - V_{TH})^2$$

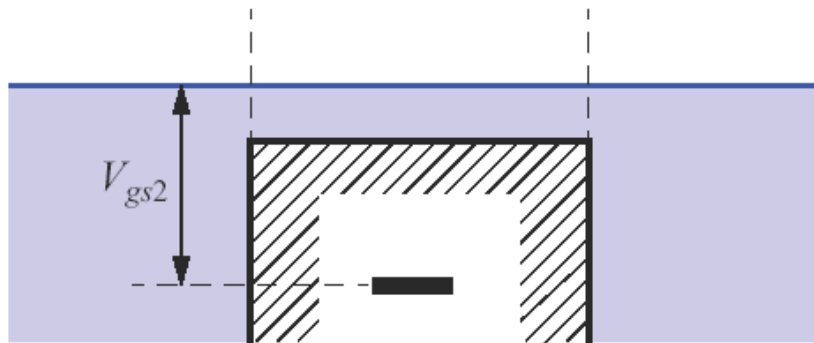


Intuition

On and off

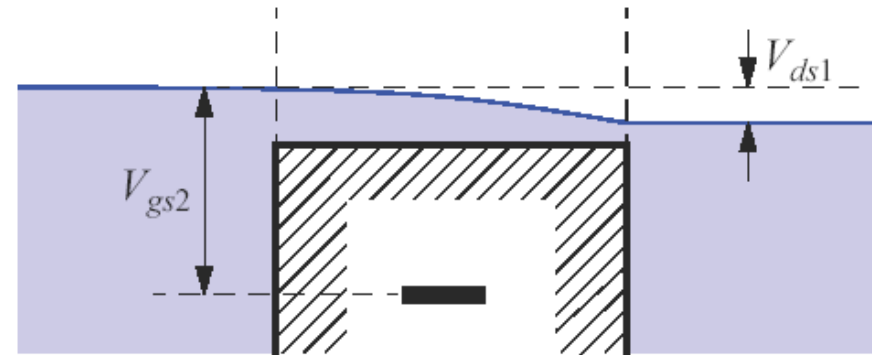


关闭 (截止)

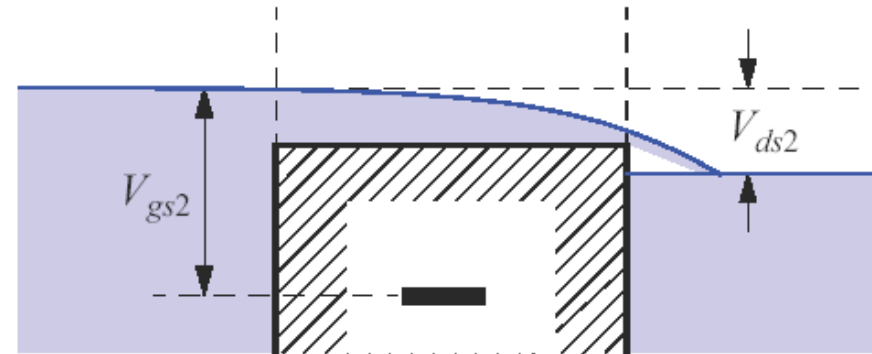


导通 (反型)

Triode and saturation



线性



饱和



Outline

- General consideration
- MOS I/V characteristics
- **MOS small signal model**
 - Notations
 - Understanding the concept of small signal
 - Transconductance of MOS
 - Ideal small signal model of MOS
- Second-order effects

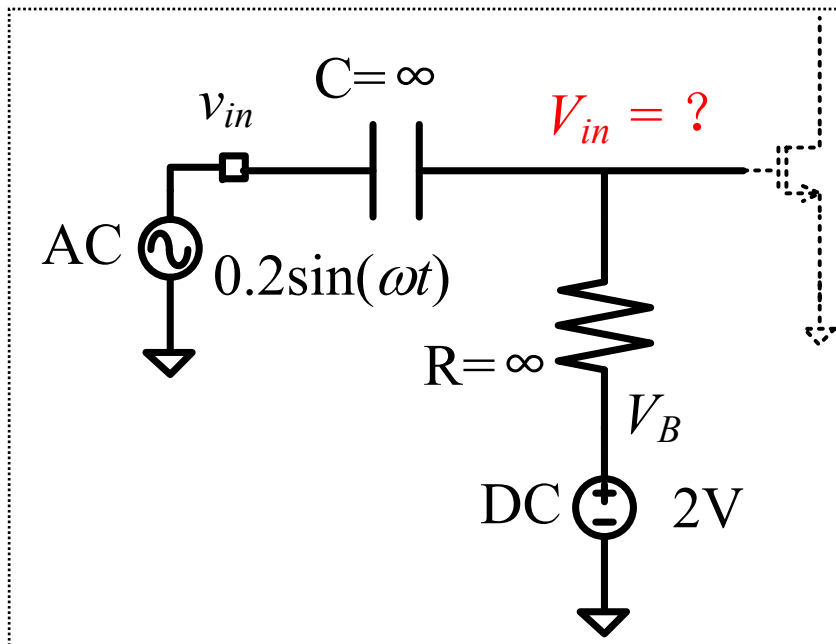
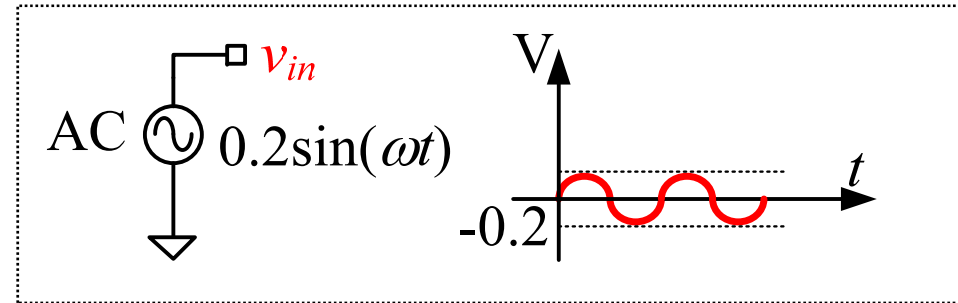
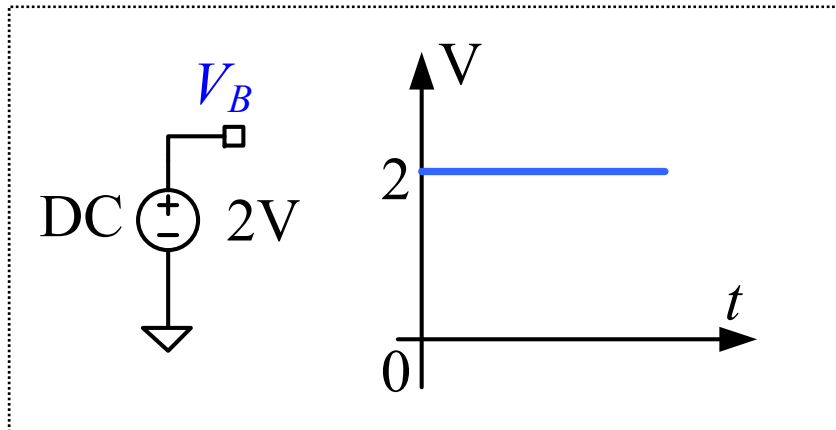


Important notations

- **DC or bias quantities: Uppercase symbol with upper case subscript**
 - DC quantities or bias voltage, currents
 - $V_{GS}, I_D, V_{DS}, V_B, \dots$
- **Small signal: Lowercase symbol with lower case subscript**
 - Incremental change of currents or voltages
 - $v_{gs}, i_d, v_{ds}, \dots$
- **Transient quantities: Uppercase symbol with lower case subscript**
 - $V_{gs} = V_{GS} + v_{gs}$

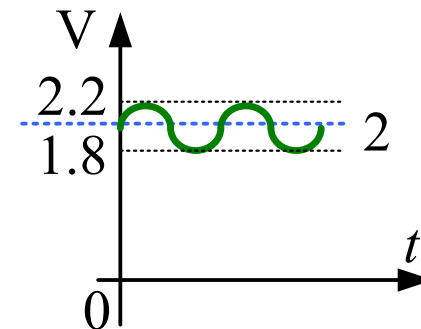


Explanation of notations



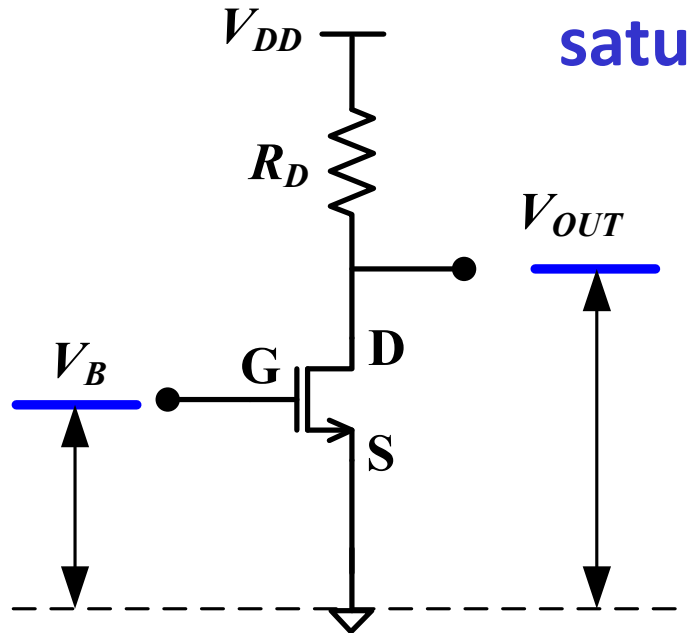
$$V_{in} = V_B + v_{in}$$

Bias (pointing to V_B)
Small Signal (pointing to v_{in})
Large Signal (pointing to V_{in})



DC or Bias analysis of a simple amplifier

Assume the MOS operates in saturation region.



$$I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_B - V_{TH})^2$$

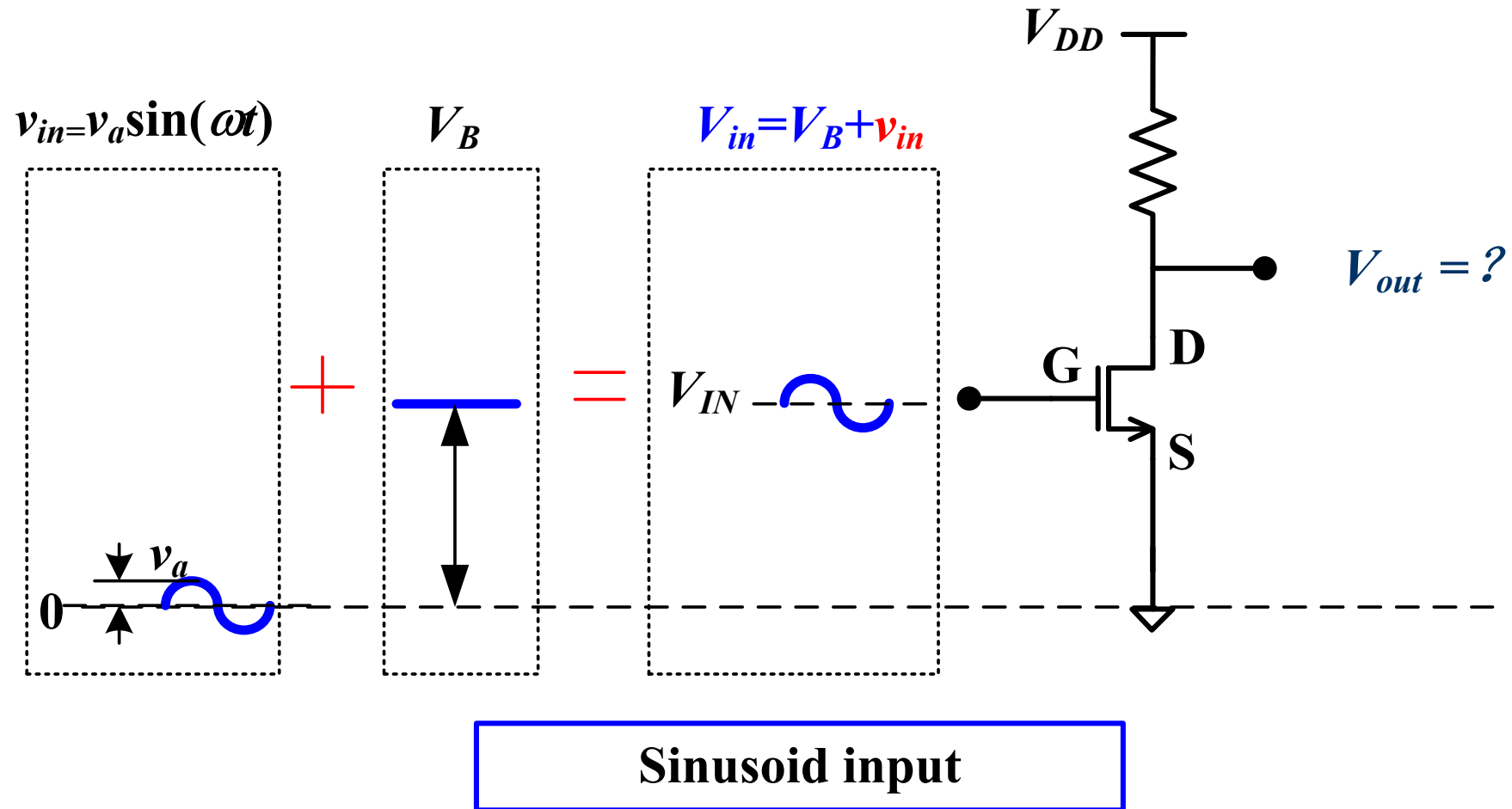
$$V_{OUT} = V_{DD} - I_D R_D$$

Can we use “ V_{OUT}/V_B ” to denote gain of the amplifier?



Transient analysis

$$V_{in} = V_B + v_{in} = V_B + v_a \sin(\omega t)$$



Quantitative analysis

$$V_{in} = V_B + v_{in} = V_B + v_a \sin(\omega t)$$

$$I_d = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{in} - V_{TH})^2 = \frac{\mu_n C_{ox}}{2} \frac{W}{L} [(V_B - V_{TH}) + v_a \sin(\omega t)]^2$$

$$= I_D + \underbrace{\left[\mu_n C_{ox} \frac{W}{L} (V_B - V_{TH}) \right] \cdot v_a \sin(\omega t)}_{i_d} + \underbrace{\frac{\mu_n C_{ox}}{2} \frac{W}{L} \cdot [v_a \sin(\omega t)]^2}_{\text{nonlinearity}}$$

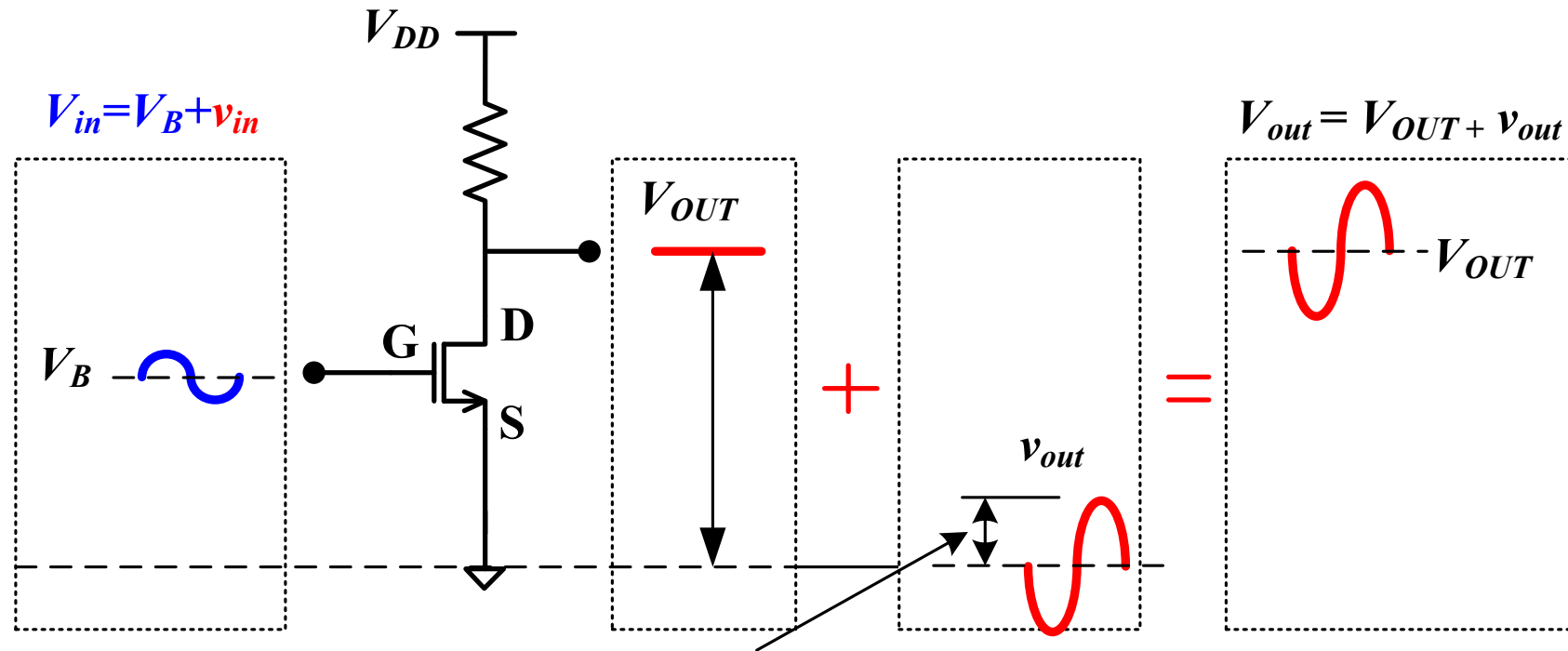
$$I_d = I_D + i_d \quad V_{out} = V_{DD} - I_d R_D = (V_{DD} - I_D R_D) - i_d R_D$$

$$= V_{OUT} + v_{out}$$

$$v_{out} = -i_d R_D = - \underbrace{\left[\mu_n C_{ox} \frac{W}{L} (V_B - V_{TH}) \right] \cdot R_D}_{\text{Amplification factor}} \cdot v_a \sin(\omega t)$$



Transient response



$$\mu_n C_{ox} \frac{W}{L} (V_{IN} - V_T) \cdot R_D \cdot v_a$$

Amplification factor

$$v_{in} = v_a \sin(\omega t)$$

$$v_{out} = -i_d R_D = -\mu_n C_{ox} \frac{W}{L} (V_B - V_{TH}) \cdot R_D \cdot v_a \sin(\omega t)$$



Small-signal model

$$v_{in} = v_a \sin(\omega t)$$

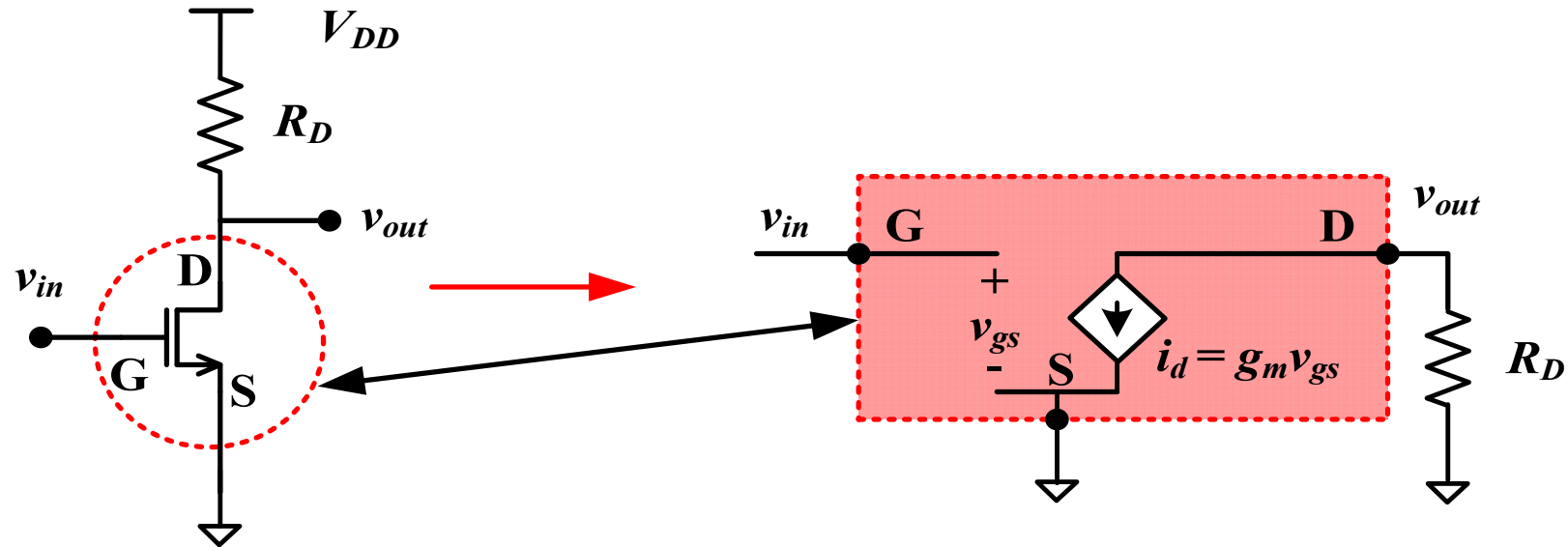
transconductance: g_m

$$v_{out} = -i_d R_D = -\boxed{\left[\mu_n C_{ox} \frac{W}{L} (V_B - V_{TH}) \right]} \cdot v_{in} \cdot R_D$$

- **Problem:** Can we analyze small signal directly?
- **Solution:** **Small signal equivalent model.**
- **decomposition of the signal transfer process:**



Basic small-signal model



- Key point: How to calculate g_m ?



g_m : Transconductance

$$g_m = \left. \frac{\partial I_D}{\partial V_{GS}} \right|_{V_{DS}=\text{constant}}$$

Control capability of V_{GS} on I_D

$$I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{GS} - V_{TH})^2$$

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})$$

$$= \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}$$

$$= \frac{2I_D}{V_{GS} - V_{TH}}$$

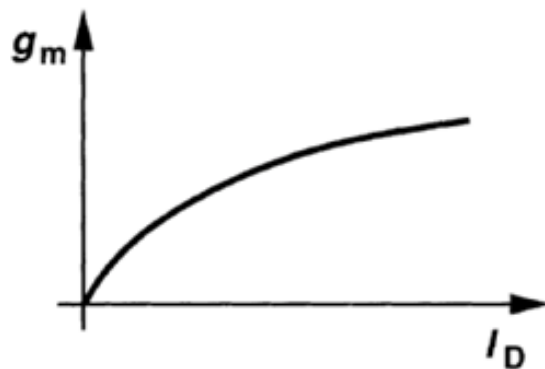
$$g_m \propto I_D \quad \text{or} \quad g_m \propto \sqrt{I_D} \quad ?$$



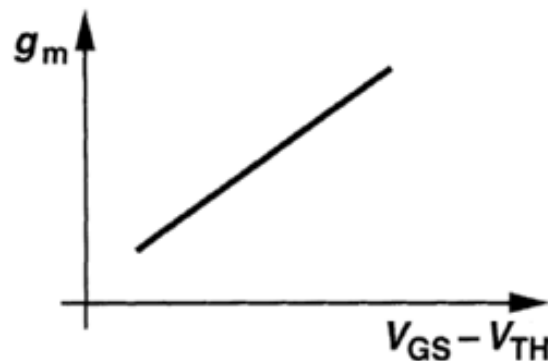
Three elements for calculating g_m

- There are **only 2** independent elements in **W/L , I_D , and $V_{GS} - V_{TH}$**
- Any one can be derived from another two

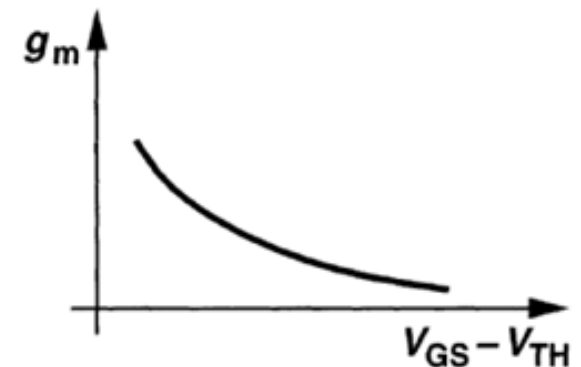
$$g_m = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_D} = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH}) = \frac{2I_D}{V_{GS} - V_{TH}}$$



W/L 不变



W/L 不变



I_D 不变



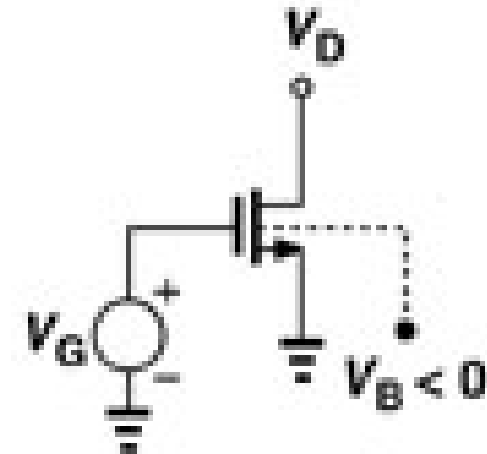
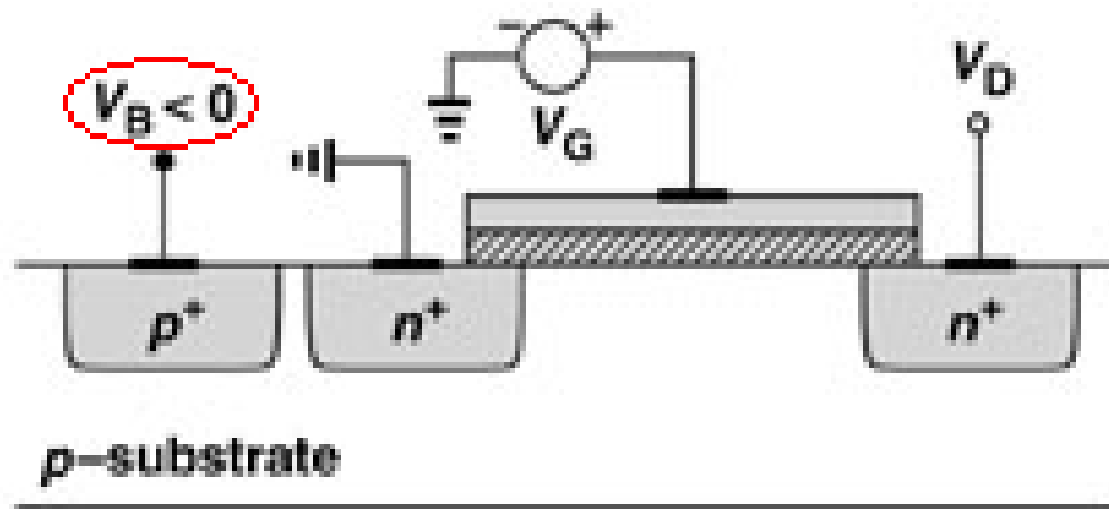
Outline

- General consideration
- MOS I/V characteristics
- MOS small signal model
- **Second-order effects**
 - Body effect
 - Channel length modulation
 - Sub-threshold conduction



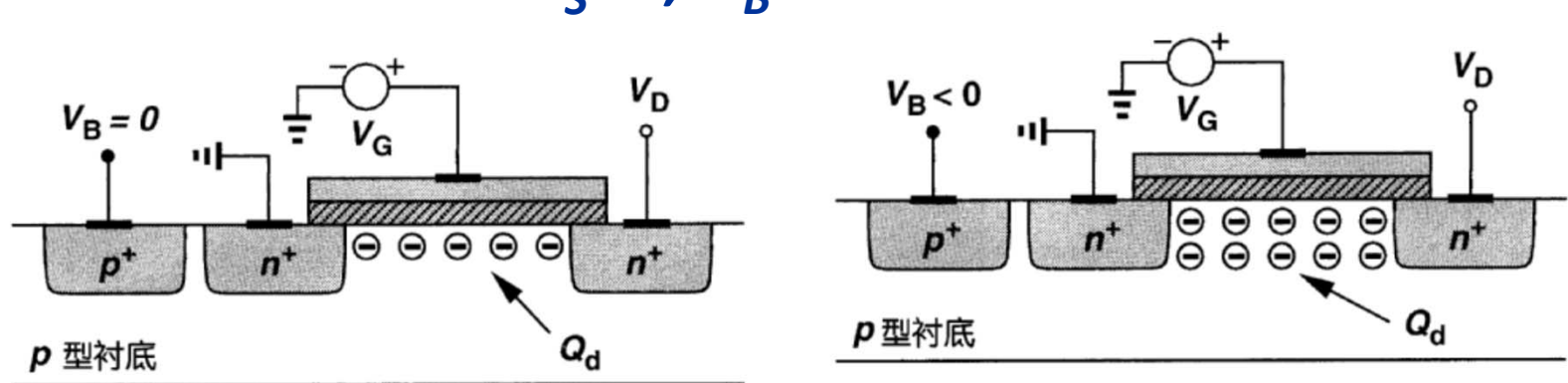
Body Effect (1)

What happens if $V_B \neq V_S$?



Influence of body effect on MOS

- Consider that $V_S=0$, $V_B < 0$



- More holes are extracted from the surface to the body electrode
- Negative charge increases, depletion region wider.
 - $\rightarrow V_B \downarrow, Q_d \uparrow$. Therefore, larger V_{GS} is needed to obtain inversion at surface. $\rightarrow V_{TH}$ increases.
 - $\rightarrow$ We call this “**body effect**” or “**backgate effect**”.



Quantitative analysis of body effect on V_{TH}

$$Q_{dep0} = \sqrt{2qN_{sub}\epsilon_{si}2\Phi_F} \quad Q_{dep} = \sqrt{2qN_{sub}\epsilon_{si}(2\Phi_F + V_{SB})}$$

$$V_{TH} = \Phi_{MS} + 2\Phi_F + \frac{Q_{dep}}{C_{ox}}$$

From Q_{dep0} to Q_{dep} ,
 $\rightarrow V_{TH} \uparrow$

$$= \Phi_{MS} + 2\Phi_F + \frac{Q_{dep0}}{C_{ox}} + \frac{Q_{dep} - Q_{dep0}}{C_{ox}}$$

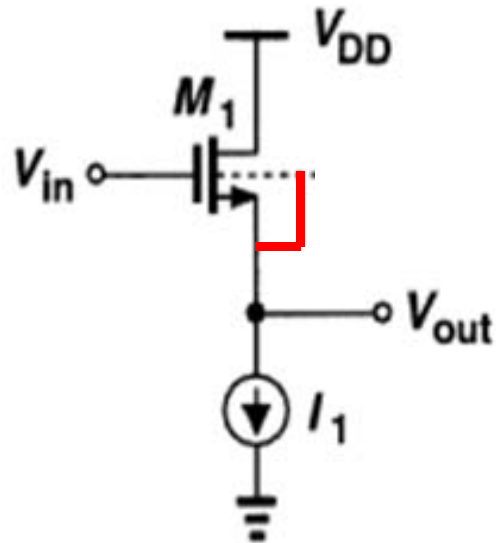
$$= V_{TH0} + \gamma \left(\sqrt{2\Phi_F + V_{SB}} - \sqrt{2\Phi_F} \right) \quad \gamma = \frac{1}{C_{ox}} \sqrt{2q\epsilon_{si}N_{sub}}$$

— γ is called **body effect coefficient**, and is provided by foundry.

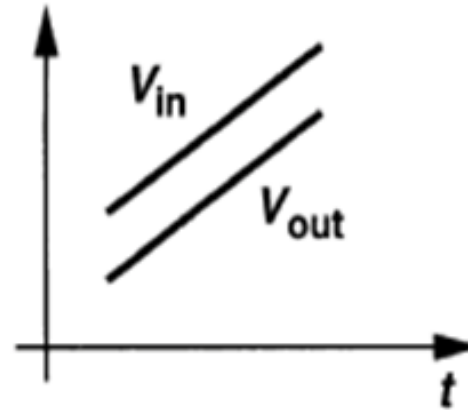
— For a 0.5um CMOS process, NMOS $\gamma = 0.4V^{1/2}$;
 PMOS $\gamma = -0.4V^{1/2}$



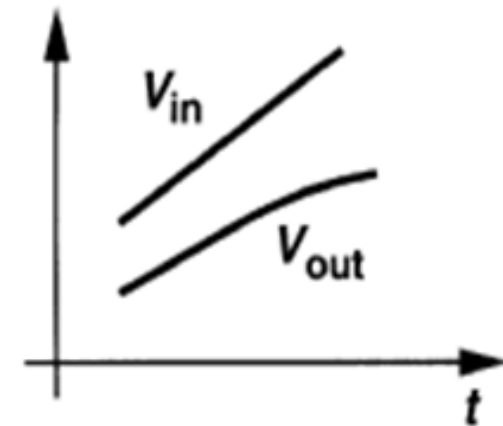
An example: influence of body effect on MOS circuit



(a)



(b)



(c)

(a) The source-bulk voltage varies with input level;

(b) input and output voltages without body effect;

(c) input and output voltages with body effect



Influence of body effect on small-signal model: g_{mb}

g_{mb} : **body transconductance**

— **Control capability of body voltage on drain current**

$$g_{mb} = \frac{\partial I_D}{\partial V_{BS}} = \mu_n C_{OX} \frac{W}{L} (V_{GS} - V_{TH}) \left(\frac{-\partial V_{TH}}{\partial V_{BS}} \right)$$

Also,

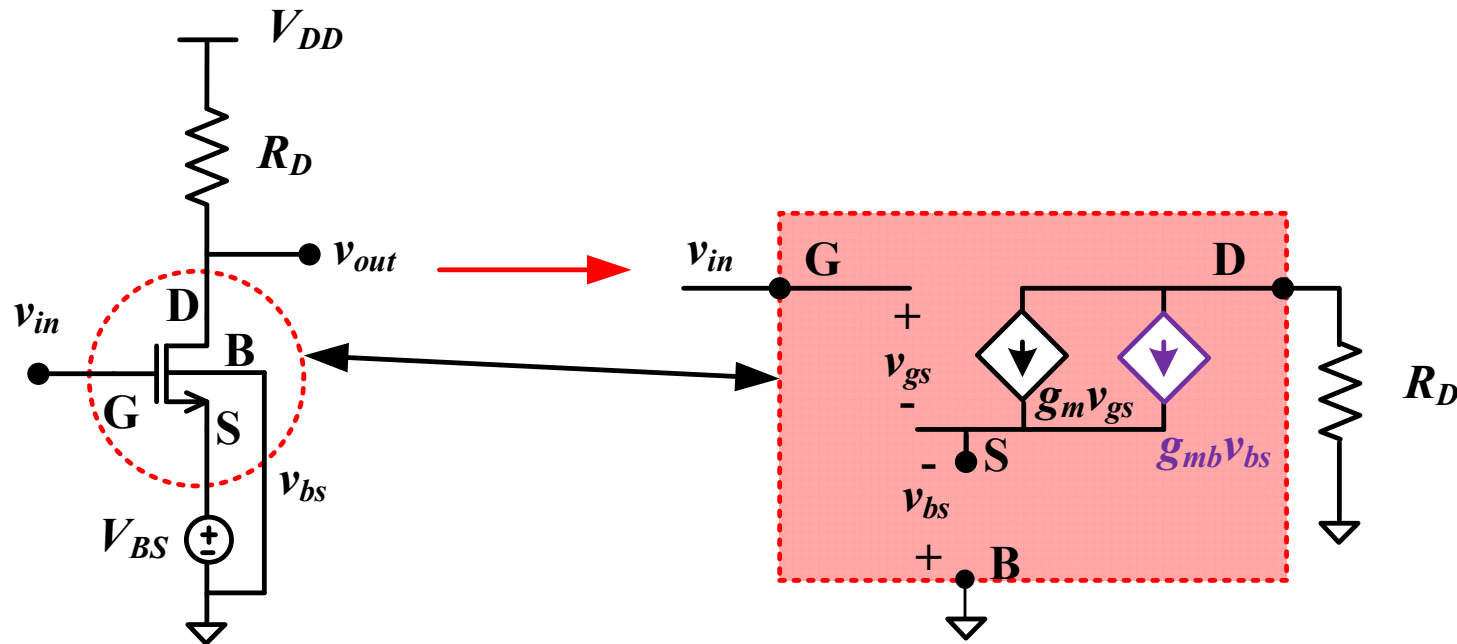
$$\frac{\partial V_{TH}}{\partial V_{BS}} = \frac{-\partial V_{TH}}{\partial V_{SB}} = -\frac{\gamma}{2} (2\Phi_F + V_{SB})^{-1/2}$$

$$g_{mb} = g_m \frac{\gamma}{2\sqrt{2\Phi_F + V_{SB}}} = \eta g_m$$

$$\eta = g_{mb} / g_m, \quad \text{typical value is } 0.1 \sim 0.3$$



Small-signal model with body effect



Body transconductance provides another freedom to control the drain current

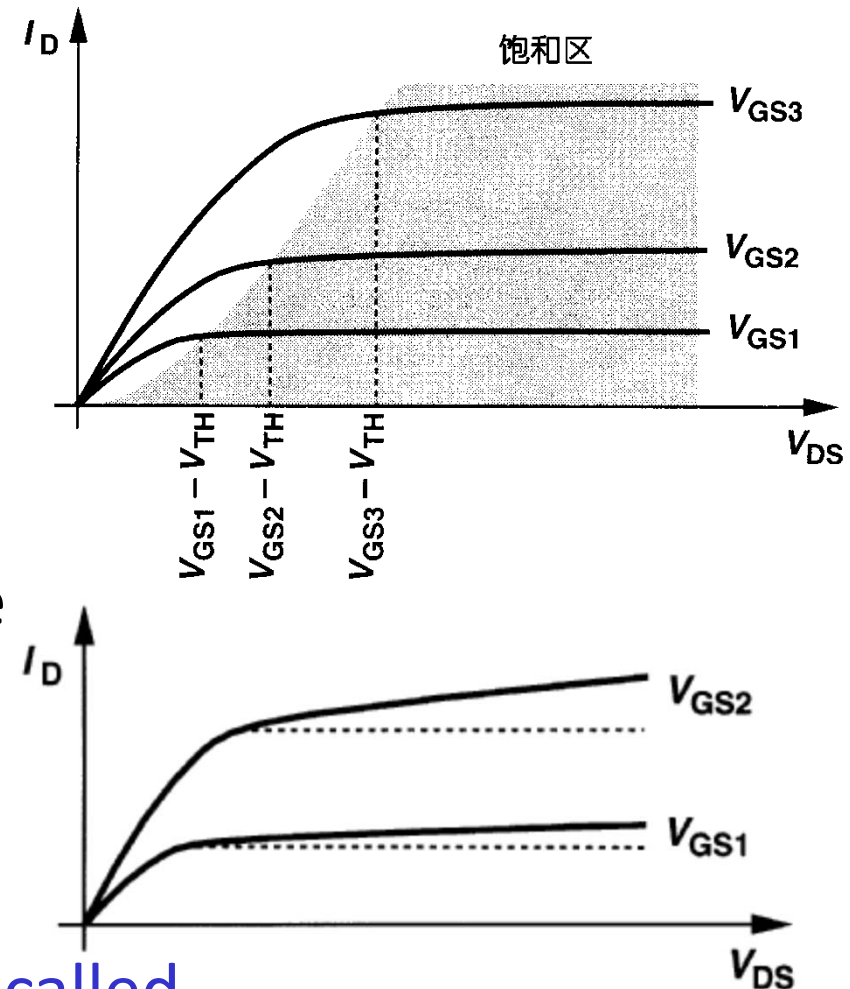
In N-well process, body of all NMOS is connected to ground.

However, the body of PMOS can be used as a control terminal



Channel length modulation effect

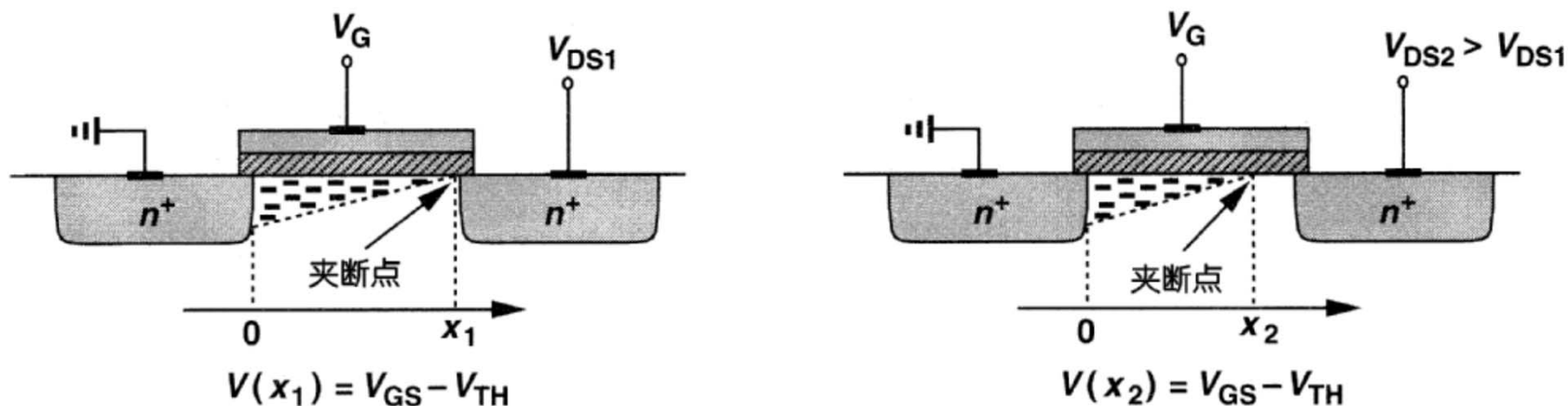
- A NMOS operated in saturation region with large enough length
 - I_D is independent of V_{DS}
 - MOS is an ideal current source
 - Infinite output resistance
- **Short channel MOS**
 - I_D is influenced by V_{DS}
 - Finite output resistance
- The influence of V_{DS} on I_D is called



Channel length modulation effect



Quantitative analysis



$$L' = L - \Delta L$$

$$\frac{1}{L'} = \frac{1}{L - \Delta L} = \frac{1}{L} \frac{1}{(1 - \Delta L / L)} \approx \frac{1}{L} \left(1 + \frac{\Delta L}{L}\right)$$

$$\text{assume } \Delta L / L = \lambda V_{DS}, \quad 1 / L' = \frac{1}{L} (1 + \lambda V_{DS})$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS})$$



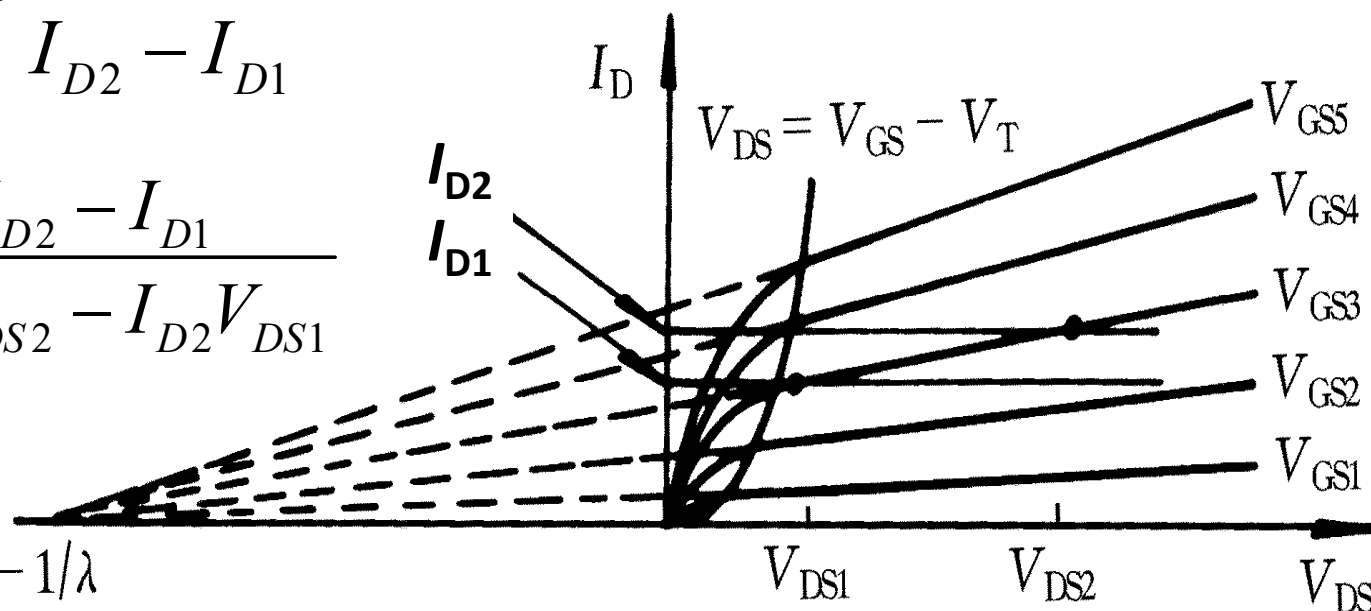
How to obtain λ ?

$$\frac{V_A + V_{DS1}}{V_{DS2} - V_{DS1}} = \frac{I_{D1}}{I_{D2} - I_{D1}}$$

$$\lambda = \frac{1}{V_A} = \frac{I_{D2} - I_{D1}}{I_{D1}V_{DS2} - I_{D2}V_{DS1}}$$

Early voltage

$$-V_A = -1/\lambda$$



$$\lambda = \frac{1}{L} \left(\frac{\Delta L}{V_{DS}} \right) \propto \frac{1}{L}$$

Different λ for different L

$$L = 0.5 \mu\text{m}, \quad \lambda \approx 0.1 \text{V}^{-1}$$

$$L = 1 \mu\text{m}, \quad \lambda \approx 0.05 \text{V}^{-1}$$



Influence of λ on small-signal model

- I_D varies with $V_{DS} \rightarrow$ non-ideal current source
- Finite output resistance r_o , or $g_{ds}=1/r_o$

$$I_D = \frac{\mu_n C_{OX}}{2} \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS})$$

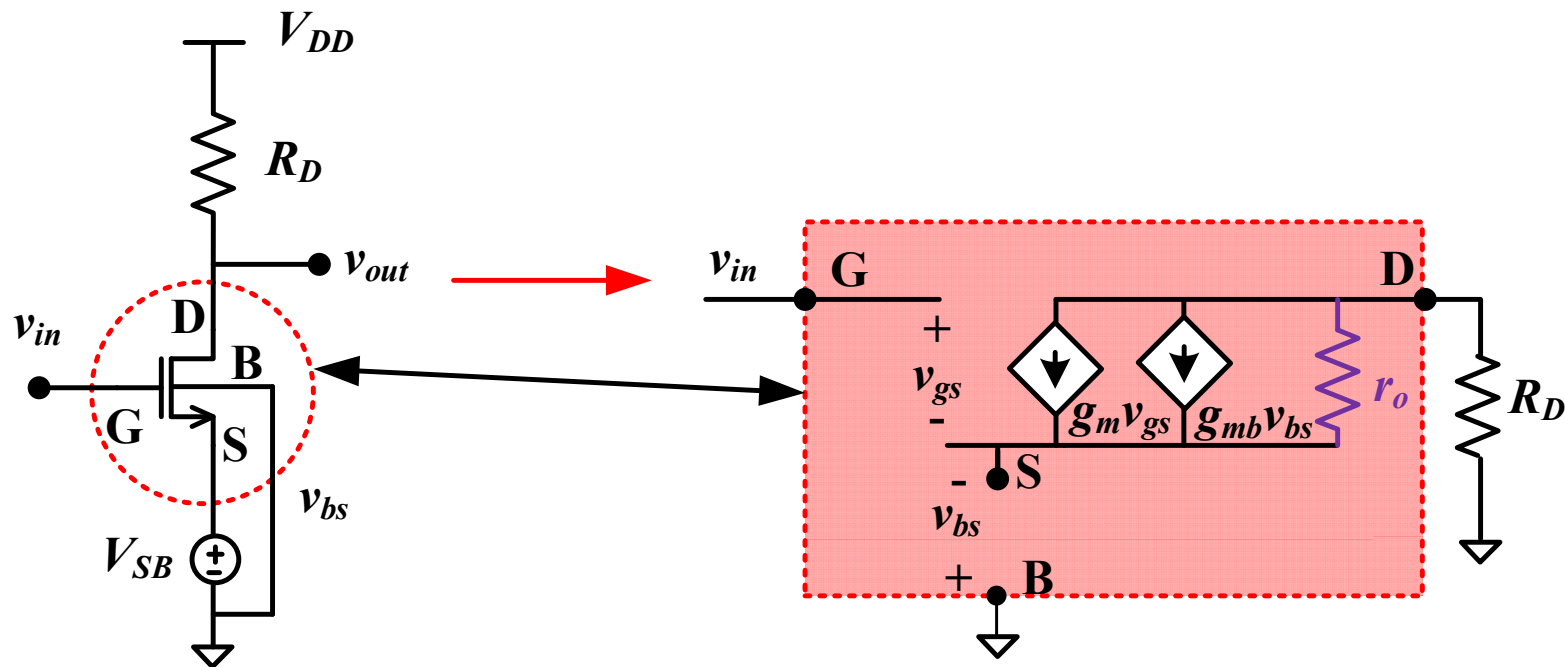
$$r_o = \frac{\partial V_{DS}}{\partial I_{DS}} = \frac{1}{\partial I_{DS} / \partial V_{DS}} = \frac{1}{\frac{1}{2} \mu_n C_{OX} \frac{W}{L} (V_{GS} - V_{TH})^2 \lambda} \approx \frac{1}{\lambda \cdot I_{DS}}$$

Important note:

MOS small-signal output resistance can be changed by adjusting bias current (I_D), or transistor length



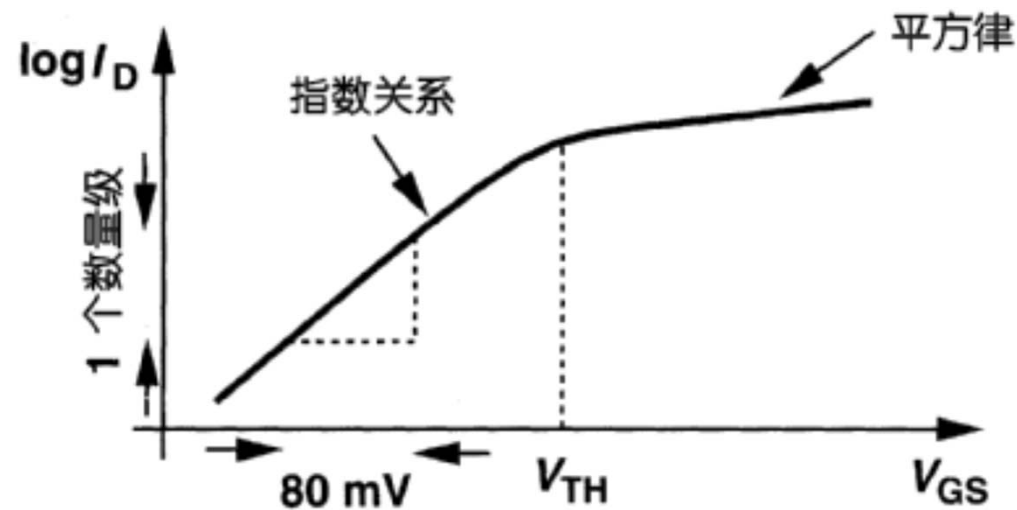
Small-signal model considering λ



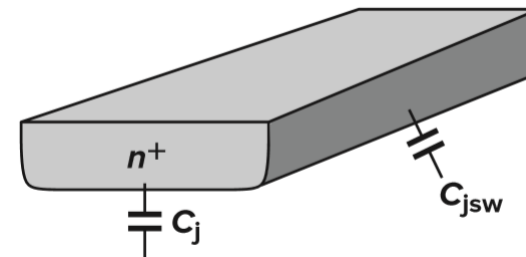
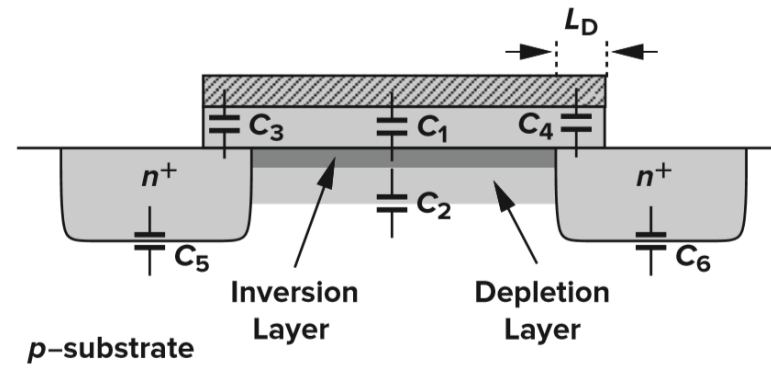
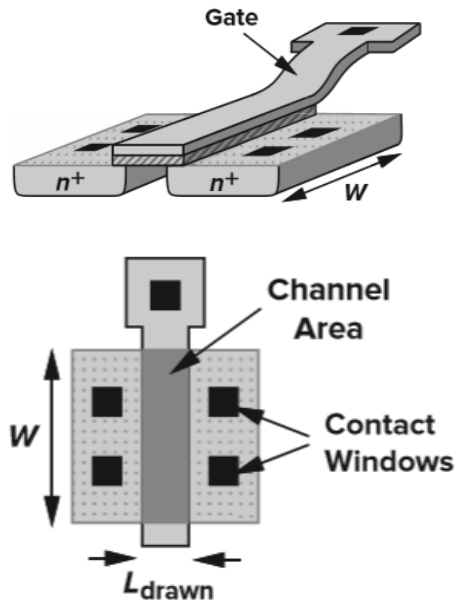
Sub-threshold Conduction

- “Weak inversion” for $V_{GS} < V_{TH}$
- Current is not zero but exhibits exponential dependence on V_{GS}
- What is the influence of this effect on real IC?

$$I_D = I_0 \exp \frac{V_{GS}}{\zeta V_T}, \quad (\zeta > 1, \quad V_{DS} > 4V_T)$$



MOS parasitic capacitance

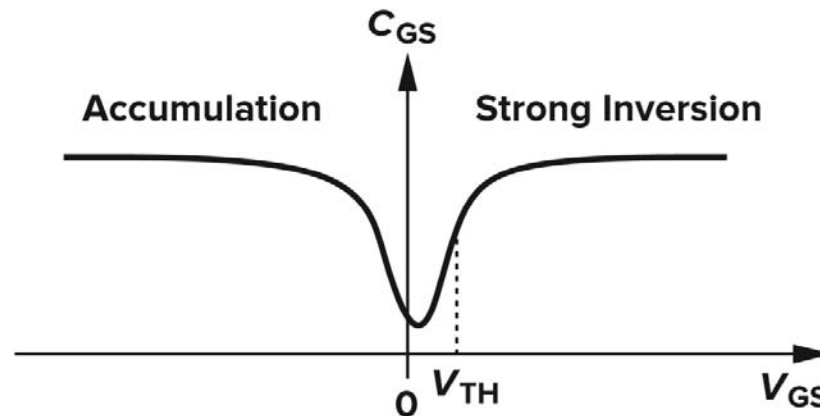
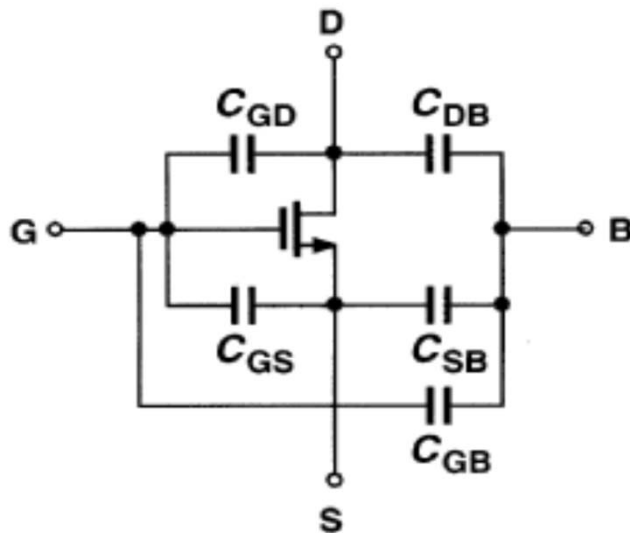


- C_1 : Oxide capacitance
- C_2 : Depletion-layer capacitance
- C_3, C_4 : Overlap capacitance
 - Overlap capacitance of unit width is C_{ov} ;
- C_5, C_6 : Junction capacitance, decomposed into C_j and C_{jsw}



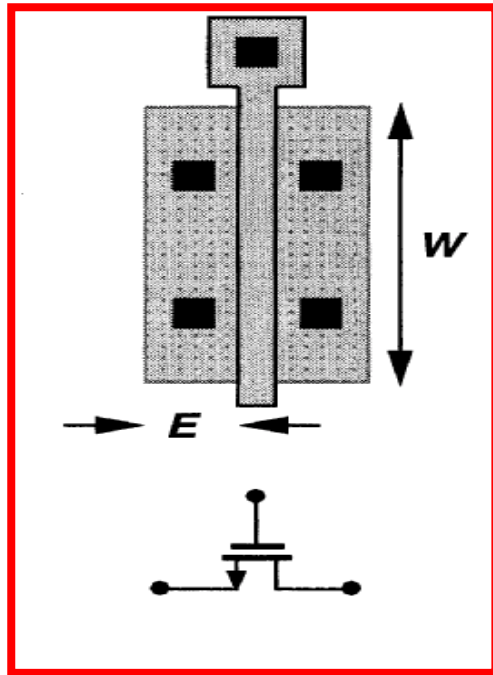
Parasitic capacitance model

- A capacitance is modeled between every two of the four terminals
 - C_{GS} and C_{GD} are more important
 - C_{GS} is a nonlinear voltage-controlled Cap.

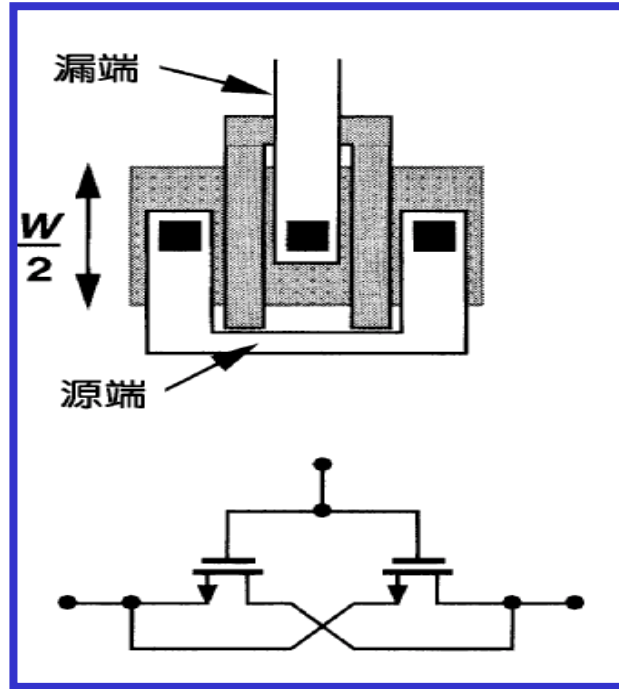


Elaborate layout design to reduce parasitic cap.

“folded or “multi-finger” structure



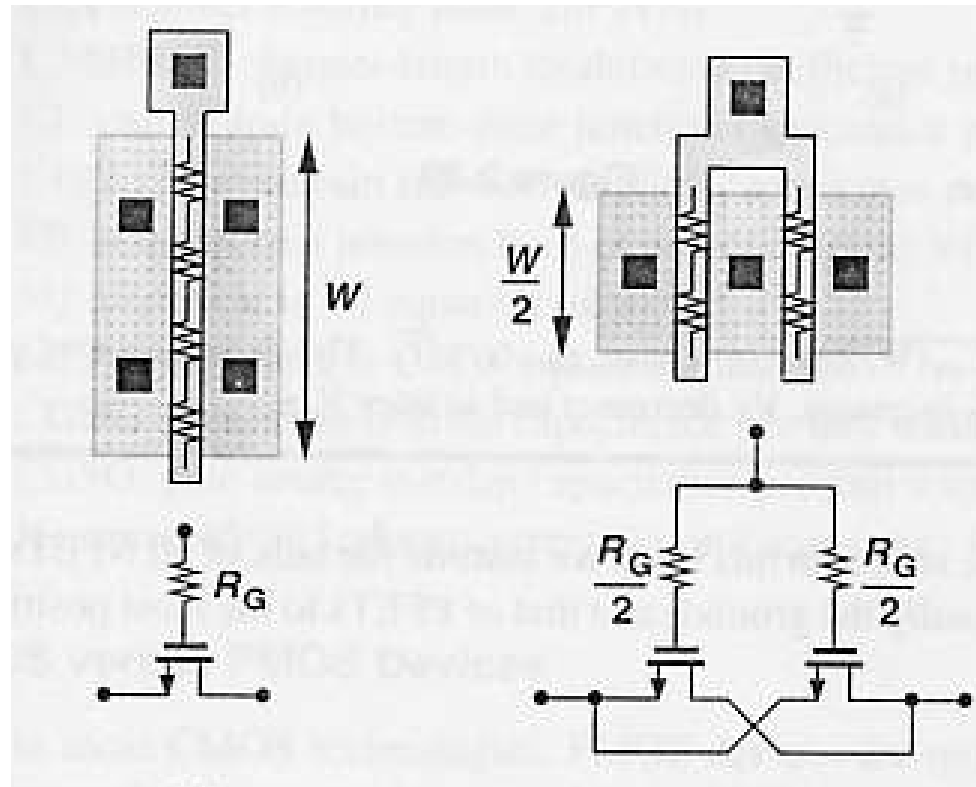
$$C_{DB} = C_{SB} = WEC_j + 2(W + E)C_{jsw}$$



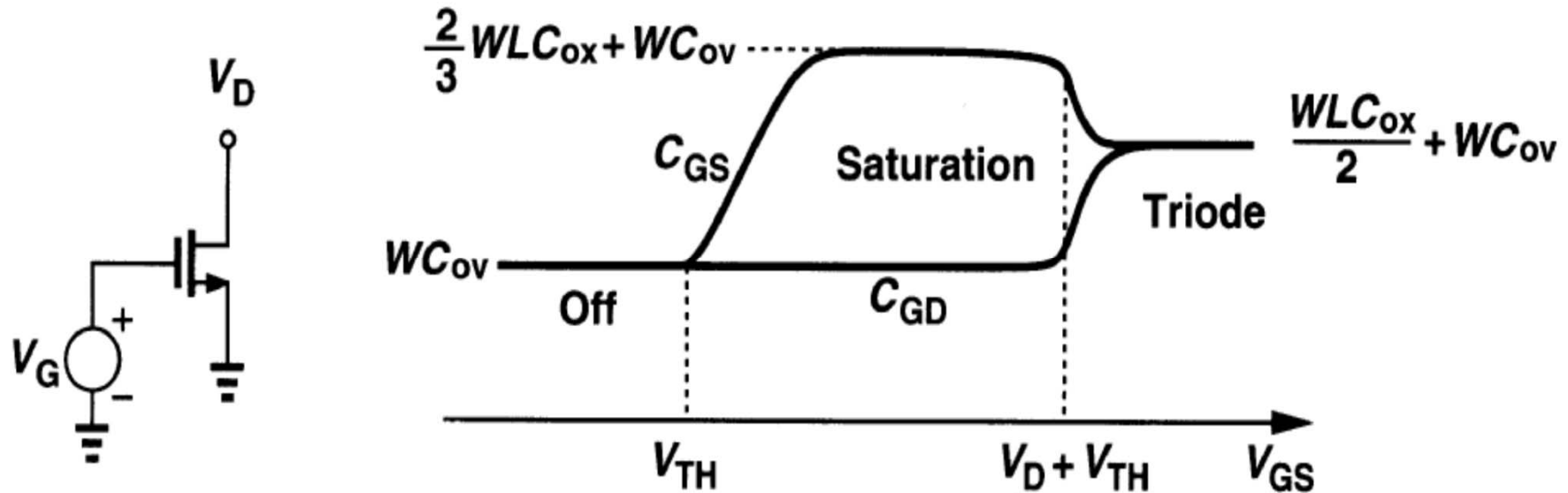
$$C_{DB} = \frac{W}{2}EC_j + 2\left(\frac{W}{2} + E\right)C_{jsw}$$



- Multi-finger structure reduces parasitic resistance



C_{GS} , C_{GD} in different operation regions

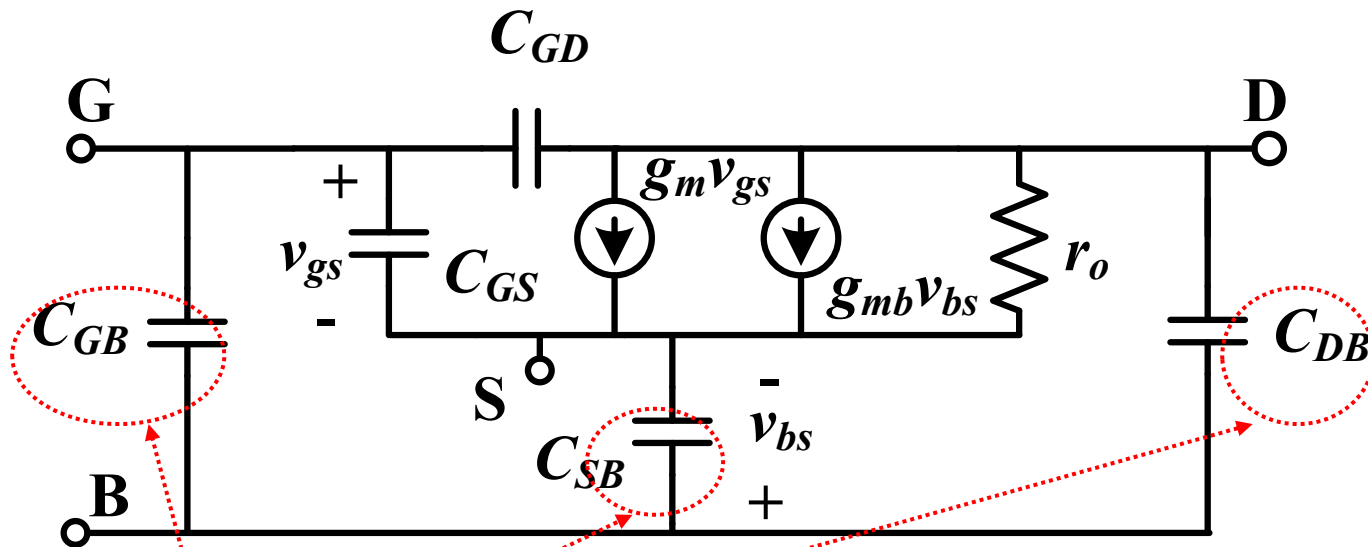


Important note:

C_{GS} is much larger than C_{GD} in saturation region



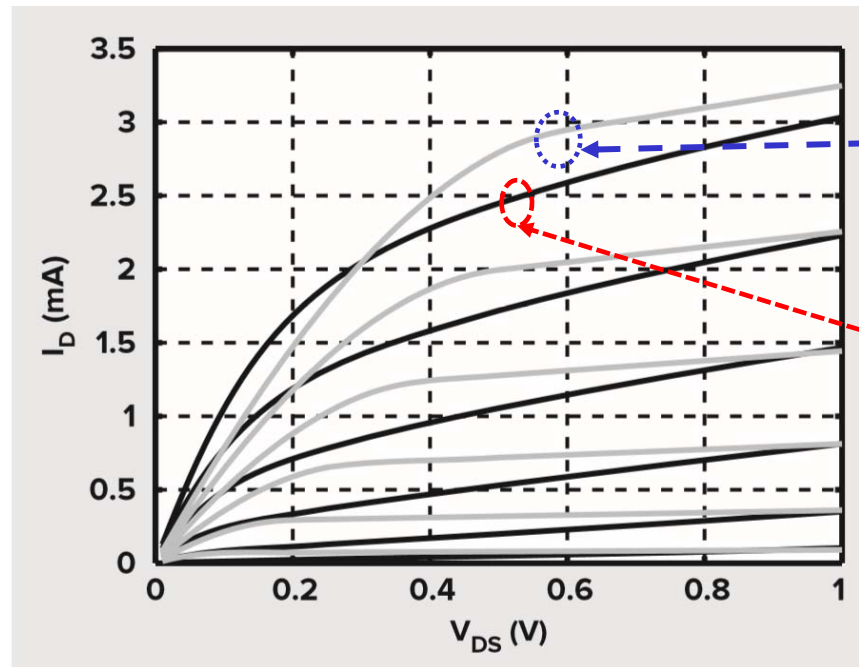
Complete MOS small-signal model



C_{GB} , C_{SB} and C_{DB} can be omitted for simplicity



I/V Curve of Nano-meter MOS



– Square-Law curves with matched W/L

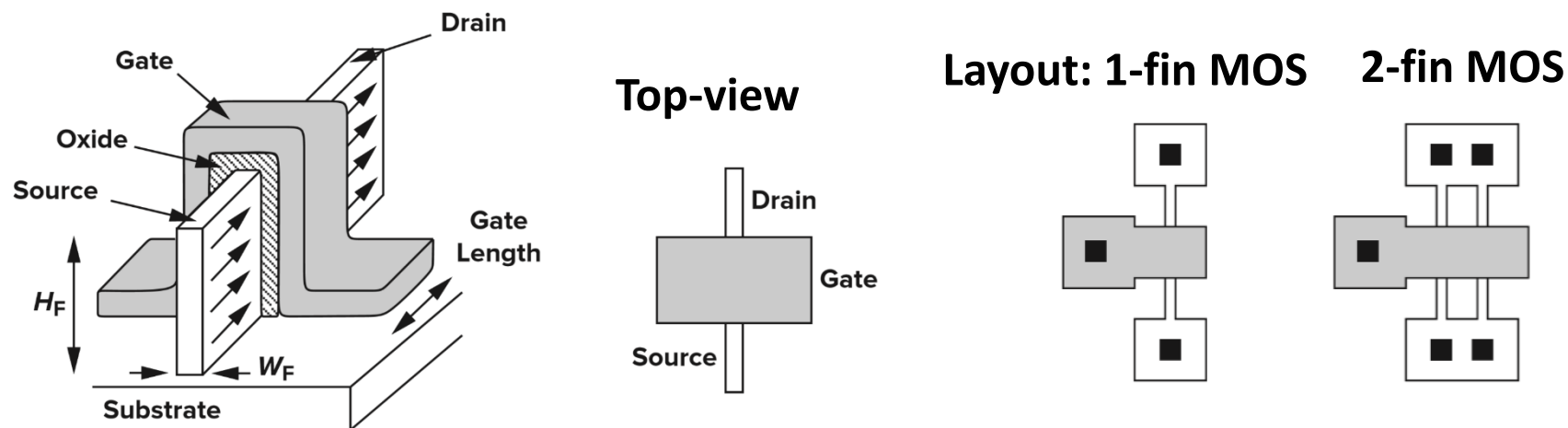
I/V curves of a 40nm Device
(W/L = 5 μ /40nm)

V_{GS} = 0.3, 0.4, ..., 0.8V

- Nano-meter MOS suffers from more imperfections
- Deviate from square-law characteristics
- Hand calculation leads to significant errors. G_m/I_D design methodology is more preferred (see Chapter. 11).
 - A reference book: *Boris Murmann, Systematic Design of Analog CMOS Circuits, 2018*



A glimpse on FinFET



- When $L_{\min} < 20\text{nm}$, 2-D device $\rightarrow$ 3-D device (FinFET)
 - FinFET has a vertical “fin”, making its I/V characteristics closer to square-law behavior
 - Our simple large-signal mode is relevant again
- $W = 2H_F + W_F$ ($W_F \approx 6\text{nm}$; $H_F \approx 50\text{nm}$). **H_F and W_F are fixed (can't be designed)**
- Practical devices are **multiple-fin structures**



summary

- General consideration
- MOS I/V characteristics
 - Threshold Voltage
 - I/V characteristics derivation
- MOS small signal model
 - Notations
 - Understanding the concept of small signal
 - Transconductance of MOS
 - Ideal small signal model of MOS
- Second-order effects
 - Body effect
 - Channel length modulation
 - Sub-threshold conduction
 - Complete small-signal model

